Being and Consciousness: Fiscal Attitudes according to HANK

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Abstract

Cross-country differences in fiscal attitudes need not reflect deep-rooted national preferences. We show this by bringing the latent political-economy dimension of HANK models to the fore and applying it to a central monetary-union issue: the national fiscal stance. Comparing household-level welfare under alternative budgetary policies reveals variation in "fiscal preferences" within countries and distinct voting majorities across countries. These majorities depend on initial public debt and the underlying shock: following a monetary tightening, the majority supports a balanced-budget policy when debt is low but deficit financing when debt is high. Following a government-spending increase, even low-debt countries support deficit financing.

Keywords: HANK, Two-country model, Political Economy, Budget deficits, Government debt, Fiscal policy, Household heterogeneity
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"Being determines consciousness"
Karl Marx[1]

1 Introduction

Attitudes toward fiscal policy differ across Europe and have been at the center of many policy debates over "excessive deficits" since the inception of the euro. These attitudes are often seen as reflecting deep-rooted national preferences (Brunnermeier et al. 2016). For instance, observers have frequently pointed out that the German language uses the same word for both debt and guilt, in an attempt to rationalize what is arguably a German obsession with debt and deficits. And while it is true that Germany, along with other "northern" members of the euro area (EA), has frequently insisted on tight budgets, in spring 2025 it voted to exempt military spending from its constitutional debt rule. This development may seem puzzling when interpreted against the backdrop of a "national psyche." Yet it can be rationalized within a materialist, that is, economic perspective.

We do so in the present paper. In terms of attitudes toward fiscal policy, we focus on the fiscal balance and distinguish between *fiscal conservatism*, which, all else equal, seeks to avoid deficits, and *fiscal liberalism*, which is more willing to tolerate them. We explain how economic circumstances ("being") shape these attitudes ("consciousness") across households, countries, and states of the world. While we do not rule out that national preferences may matter at times, our analysis seeks "to push the rationality postulate as far as it will go" (Bernanke 1983).

To this end, we develop a two-country heterogeneous-agent New Keynesian (HANK) model and bring its latent political-economy dimension to the fore by deriving majority support from household-level welfare rankings over alternative fiscal policies. To capture key features of the EA, countries differ along two dimensions: government debt is higher in *South* than in *North*, and the wealth distributions differ in line with the data.

We study fiscal attitudes in the context of a monetary tightening. Fiscal conservatism responds to its adverse budgetary consequences by raising taxes immediately, generating a deeper but shorter recession, whereas fiscal liberalism issues debt and spreads the adjustment over time. "Fiscal preferences" vary substantially within each country, yet aggregate into different majorities across countries. In *North*, the majority supports fiscal conservatism, whereas in *South*, it supports fiscal liberalism. This difference is driven primarily by fiscal exposure: higher initial government debt in *South* makes the short-run adjustment under fiscal conservatism more severe. These majorities are not fixed, however. Following an increase in government spending, even the majority in *North* supports fiscal liberalism. Moreover, we find that the household-level effects of alternative fiscal policies predicted by the model can rationalize voter preferences revealed in survey data.

[1]The quote is a commonly used shorthand for: "It is not the consciousness of men that determines their being, but, on the contrary, their social being that determines their consciousness." (Marx 1859).

More in detail, we develop a two-country heterogeneous agent New Keynesian model: HANK[2]. The model embeds the HANK structure of Bayer et al. (2024a) in an otherwise standard two-country framework, as in Corsetti et al. (2012). Countries specialize in differentiated goods used for consumption and investment in both economies, prices are sticky, monetary policy is common, and financial markets are segmented. Within each country, households face uninsurable idiosyncratic income risk and borrowing constraints. They save in illiquid capital, which yields a higher return but provides limited insurance and is not traded internationally, and in liquid government bonds, which are traded within and across countries. Fiscal policy is set nationally through a distortionary income tax that adjusts more or less aggressively to stabilize public debt.

The key distinction between the two countries concerns how households insure against idiosyncratic income risk. The literature has established that countries may adopt different strategies to cope with market incompleteness: they may rely on direct public insurance, or households may rely more heavily on government bonds for self-insurance. The welfare state thus shapes the steady-state demand for public debt, as stronger minimum-income protection reduces precautionary saving in government bonds (Hubbard et al. 1995; Aiyagari and McGrattan 1998; Mian et al. 2021). We incorporate this mechanism into our model to microfound cross-country differences in steady-state debt levels. For our results below, however, what matters primarily is that debt levels differ across countries, rather than the precise source of this difference.

In our baseline calibration, *South* has high public debt and provides no minimum-income benefits. We calibrate it to capture key features of the Italian economy. Italy is not only the largest southern EA economy but also lacked a permanent minimum-income scheme until recently. Specifically, we choose model parameters, including those governing household income risk, to match key moments of Italy's income and wealth distributions, its aggregate capital stock, and a public-debt-to-GDP ratio of 132%. We then construct *North* as a twin economy that shares the same preference and technology parameters but features lower public debt and a minimum-income scheme calibrated to capture evidence from Germany. We then choose the debt level so as to keep effective consumption insurance at the Euler equation—and hence the steady-state real interest rate—unchanged (Werning 2015). Because minimum-income benefits reduce the need for precautionary saving in government bonds, households in *North*, when facing the *South* interest rate, demand public debt amounting to only 95% of GDP. We assume that the government in *North* issues precisely this amount, ensuring a zero net foreign asset position. At the same time, weaker incentives for low-wealth households to accumulate assets increase wealth inequality; see also Pham-Dào (2016). Strikingly, the resulting wealth distributions in *North* and *South* align closely with the actual differences between Germany and Italy.[2]

[2]To ensure robustness, we recalibrate *North* to directly match the German data on income and wealth distributions, capital holdings, and government debt. Our main results remain qualitatively unchanged.

To study fiscal attitudes over the business cycle, we consider a monetary contraction and—to set the stage—first focus on the limiting case of a closed economy, comparing outcomes in *North* and *South* in isolation. A monetary contraction leads to a fiscal shortfall: interest rates rise and tax revenues decline. The conservative fiscal policy responds by raising taxes immediately to keep the government budget balanced. In contrast, a liberal fiscal policy buffers the shock by issuing new debt. The policy alternatives entail a trade-off: raising taxes to balance the budget causes a deeper but shorter recession, as stressed by Kaplan et al. (2018), whereas smoothing tax distortions through debt issuance leads to a milder but more persistent recession, potentially increasing cumulative output losses. Yet, while immediate tax hikes can avoid prolonged distortions to economic activity, they generate cost-push inflation that weakens the disinflationary effects of a monetary tightening. This puts upward pressure on real interest rates, which can make fiscal adjustment more costly—especially in *South*, where higher initial debt implies that a given increase in real rates translates into higher debt-service costs.

Our focus is on how the costs and benefits of alternative fiscal policies are distributed across households. We assess these effects using a one-sided welfare measure, defined as the consumption-equivalent variation that makes a household as well off in the presence of a shock as in its absence. By comparing welfare across policies at the household level and focusing on the median voter, we infer "country preferences" over fiscal policy at business-cycle frequency. In terms of average welfare, fiscal liberalism performs better in both countries, with a larger advantage in *South* because immediate tax adjustments are more costly there. Average welfare, however, masks substantial heterogeneity: low-wealth households incur particularly large welfare losses under fiscal conservatism because they are less able to absorb the deeper initial recession, whereas for most households the welfare differences between the two policies are much smaller. As a result, the average-welfare ranking of policies does not always coincide with the preference of the median voter. And while the distribution of welfare effects is broadly similar across countries, the resulting national majorities differ. About 60% of households in *North* are better off under conservative fiscal policy. In *South*, by contrast, only 44% of households "support" conservative fiscal policy. Notably, the same relative preference emerges after a monetary expansion, although in that case fiscal conservatism entails running a budget surplus and reducing debt, while fiscal liberalism maintains a balanced budget.

We find that the higher support for conservative fiscal policy in *North* is driven primarily by its more favorable macroeconomic adjustment due to a lower initial public debt level. In a counterfactual exercise, we impose the general-equilibrium adjustment paths in *North* on households in *South* while keeping their wealth distribution and other household characteristics fixed. This accounts for most of the cross-country difference in support for conservative fiscal policy.

We then show that attitudes toward fiscal policy depend on the nature of the underlying

shock. To illustrate this, we consider a government spending shock. In this case, there is an overwhelming majority in favor of liberal fiscal policy in both, *South* and *North*. Even in *North*, only around 10% of households opt for fiscal conservatism. Intuitively, balancing the budget would require a substantial tax increase in the face of higher government spending—largely independent of the initial debt level, since it primarily finances the additional spending. These strong front-loaded tax hikes put upward pressure on real interest rates, making conservative fiscal policy more costly. Hence, in the face of temporarily higher spending, there is a case for tax smoothing, as in Barro (1979), even though we abstract from convex tax distortions.

Also in this case, however, the welfare costs of alternative fiscal policies vary substantially across households. We confront this model prediction with new survey evidence from German households collected in the spring of 2025. The survey asks respondents whether they support the shift toward a more liberal fiscal policy to finance higher government spending that occurred at that time. As a proof of concept, we show that the heterogeneous economic effects of alternative policies predicted by our model within *North* can rationalize the voter preferences revealed in the survey data.

Finally, we revisit fiscal policy attitudes in the context of the fully-fledged, two-country model of the monetary union. A first result is that the qualitative country-level pattern remains unchanged—assuming, counterfactually, that such policies are put to a union-wide vote. As before, the median voter in *North* prefers a conservative fiscal policy, while the median voter in *South* favors a liberal one. Quantitatively, however, terms-of-trade effects amplify the cross-country differences in fiscal attitudes, as we discuss below. Aggregating across countries, we find a slim union-wide majority in favor of liberal fiscal policy, conditional on a monetary contraction. However, while the EA has put in place common fiscal rules, fiscal policy is still largely determined at the national level. To account for this, we consider voting on fiscal policy within each country, conditional on fiscal policy choices in the other country. Two observations are noteworthy. First, while *North* is still fiscally more conservative, neither country prefers to adopt a conservative fiscal policy unilaterally. Second, if households were to vote on the other country's fiscal policy, a large majority in each country would prefer the other country to run a conservative policy, regardless of their own country's fiscal policy. Taken together, these patterns suggest a systematic bias toward adopting a more liberal fiscal policy than the partner country.

The remainder of the paper is organized as follows. We conclude this introduction by relating our paper to the existing literature. Section 2 presents our HANK^2 model. Section 3 explains our calibration which specifies *North* and *South*. Section 4 presents and discusses the results for the limiting case of a closed economy and relates the model predictions for *North* to the survey evidence from Germany. Section 5 shows results for the full monetary-union framework. A final section offers some conclusions.

Related literature. Our paper relates to four strands of the literature. First, there is earlier work on monetary-policy tradeoffs in heterogeneous currency unions (Mundell 1961; Alesina and Barro 2002; Benigno 2004; Benigno and López-Salido 2006; Enders et al. 2013; Galí and Monacelli 2016) and related work which analyzes how the income distribution impinges on the political economy of fiscal unions (Bolton and Roland 1997; Alesina and Perotti 1998).

Second, a distinct strand of the literature focused on monetary and fiscal interactions in currency unions (Kenen 1969; Beetsma and Jensen 2005; Gali and Monacelli 2008; Farhi and Werning 2017; Hettig and Müller 2018; Camous and Cooper 2019) and, more specifically, the role of national debt levels for monetary policy in a currency union (Beetsma and Uhlig 1999; Sims 1999; Corsetti et al. 2014).

Third, there is work which studies household heterogeneity and incomplete financial markets within countries in an open-economy context (Mendoza et al. 2009; Ferra et al. 2021). A number of studies also extend the HANK framework put forward by Kaplan et al. (2018) and Bayer et al. (2019) to an open-economy context. Small-open economy versions are used to study the distributional implications of sudden stops, exchange rate shocks, and monetary and fiscal policy (de Ferra et al. 2020; Guo et al. 2020; Auclert et al. 2021; Druedahl et al. 2025; Acharya and Challe 2025) and the impact of foreign shocks (Zhou 2021; Druedahl et al. 2022; Oskolkov 2023). Recent work has also examined fiscal policy in multi-country HANK models, both within and outside monetary unions (Aggarwal et al. 2023; Bellifemine et al. 2022, 2025; Chen et al. 2025). Our study is unique in revealing the latent political-economy dimension of HANK models and applying it to core issues in currency unions. In related work, we use this framework to study the distributional impact of asymmetric shocks across countries in the EA (Bayer et al. 2024b), as well as of fiscal responses to an energy shock, both within and across borders (Bayer et al. 2025).

Finally, our work relates to optimal policy in heterogeneous agent economies. Based on more tractable models (Challe and Ragot 2016; Bilbiie 2025), some authors are reassessing what an "optimal" fiscal or monetary policy would look like (Bhandari et al. 2021; Le Grand et al. 2024; Acharya et al. 2023; Gornemann et al. 2021). In a heterogeneous agent framework, however, it seems natural to consider a political economy question: What rationalizes observed policy choices in a democracy? This and related questions have some tradition in public finance and the literature on optimal taxation, see, for instance, Meltzer and Richard (1981), Acemoglu et al. (2008, 2010), Brett and Weymark (2017), Bierbrauer and Boyer (2016), Bierbrauer et al. (2021), and Bierbrauer et al. (2022) and recent examples in a dynamic standard incomplete markets setting (Brendler 2020; Le Grand et al. 2025). Our paper is the first attempt to rationalize observed policy choices based on "voter" preferences within the HANK framework by exploring the distributional impact of fiscal policies over the business cycle; for related earlier work in a real business cycle context, see Bachmann and Bai (2013) and Bachmann et al. (2020).

2 Model

We model a monetary union of two countries, *South* and *North*. Markets are incomplete and households face idiosyncratic risks, but they can self-insure. As a result households become heterogeneous in income and wealth. Prices and wages are sticky as is standard in the New-Keynesian literature. The set-up in each country is closely related to the HANK model in Bayer et al. (2024a). Each country consists of a firm sector and a household sector, respectively. The firm sector of each country comprises (a) perfectly competitive intermediate goods producers who rent out labor services and capital on a national labor and a national capital market, respectively; (b) final goods producers that face monopolistic competition producing differentiated final goods out of homogeneous domestic intermediate inputs; (c) a representative consumption good bundler bundling domestic and imported foreign final goods to consumption goods; (d) producers of capital goods that turn consumption goods into capital subject to adjustment costs; (e) labor packers that produce labor services combining differentiated labor from (f) unions that differentiate raw labor rented out from households. Price setting for the final goods as well as wage setting by unions is subject to a pricing friction à la Calvo (1983). We assume that only final goods can be traded between both countries.

Within each country, the household distribution is normalized to unit mass. When aggregating across the monetary union, we weight South and North by their population shares $n_S \in (0,1)$ and $n_N = 1 - n_S$, respectively. We index *South* with S and *North* with N. Households in both countries consume a bundle which consists of home and foreign final goods bundles. Households earn income from supplying (raw) labor and capital to the national labor and the national capital markets and from owning their national firm sector, absorbing all its rents that stem from the market power of unions and final good producers, and decreasing returns to scale in capital goods production.

The government sector runs a common monetary authority and national fiscal authorities. The respective fiscal authority levies taxes on labor income and distributed profits, issues government bonds, and adjusts taxes to stabilize its debt in the long run. In addition, they run a targeted transfer system. The monetary authority sets the nominal interest rate on both government bonds according to a Taylor rule. We assume that the set-up of both countries is symmetric, while we allow for different calibrations of the transfer system in each country. In what follows, we describe the households and the firms sector only from the perspective of the *South*. We denote foreign variables from *South*'s perspective with an asterisk *.

2.1 Households

The household sector is subdivided into two types of agents: workers and entrepreneurs. The transition between both types is stochastic. Both rent out physical capital, but

only workers supply labor. The efficiency of a worker's labor evolves randomly exposing worker-households to labor-income risk. Entrepreneurs do not work but earn all pure rents in our economy except for the rents of unions which are equally distributed across workers. All households self-insure against the income risks they face by saving in a liquid nominal asset (bonds) and a less liquid asset (capital). Trading illiquid assets is subject to random participation in the capital market. To be specific, households are ex-ante identical, indexed by i. They are infinitely lived, have time-separable preferences with time discount factor β, and derive felicity from consumption c_{it} and leisure. They obtain income from supplying labor, l_{it}, from renting out capital, k_{it}, and from earning interest on bonds, b_{it}, and potentially from profits or union transfers. Households pay taxes on labor and profit income.

2.1.1 Productivity, labor supply, and labor income

A household's gross labor income $w_t l_{it} h_{it}$ is composed of the aggregate wage rate on raw labor, w_t, the household's hours worked, l_{it}, and its idiosyncratic labor productivity, h_{it}. We assume that productivity evolves according to a log-AR(1) process with time-varying volatility and a fixed probability of transition between the worker and the entrepreneur state:

$$\tilde{h}_{it} = \begin{cases} \exp(\rho_h \log \tilde{h}_{it-1} + \epsilon^h_{it}) & \text{with probability } 1-\zeta \text{ if } h_{it-1} \neq 0, \\ 1 & \text{with probability } \iota \text{ if } h_{it-1} = 0, \\ 0 & \text{else.} \end{cases} \tag{1}$$

with individual productivity $h_{it} = \frac{\tilde{h}_{it}}{\int \tilde{h}_{it} di}$ such that $\tilde{h}_{it}$ is scaled by its cross-sectional average, $\tilde{h}_{it} di$, to make sure that average worker productivity is constant. The shocks ϵ^h_{it} to productivity are normally distributed with variance $\sigma^2_{h,t}$. With probability ζ households become entrepreneurs ($h = 0$). With probability ι an entrepreneur returns to the labor force with median productivity. In our baseline specification, an entrepreneur obtains a share of the pure rents (aside from union rents), Π^F_t, in the economy (from monopolistic competition in the goods sector and the creation of capital). We assume that the claim to the pure rent cannot be traded as an asset. Union rents, Π^U_t are distributed lump sum across workers, leading to labor-income compression. For tractability, we assume union profits to be taxed at a fixed rate independent of the recipient's labor income.[3]

With respect to leisure and consumption, households have Greenwood et al. (1988)

[3]This modeling strategy serves two purposes. First and foremost, it generally solves the problem of the allocation of pure rents without distorting factor returns and without introducing another tradable asset. Second, we use the entrepreneur state in particular – a transitory state in which incomes are very high – to match the income and wealth distribution following the idea by Castaneda et al. (1998). The entrepreneur state does not change the asset returns or investment opportunities available to households.

(GHH) preferences and maximize the discounted sum of felicity:

$$E_0 \max_{\{c_{it},l_{it}\}} \sum_{t=0}^{\infty} \beta^t u[c_{it} - G(h_{it}, l_{it})] \tag{2}$$

The maximization is subject to the budget constraints described further below. The felicity function u exhibits a constant relative risk aversion (CRRA) with risk aversion parameter $\xi > 0$,

$$u(x_{it}) = \frac{1}{1-\xi} x_{it}^{1-\xi}, \tag{3}$$

where $x_{it} = c_{it} - G(h_{it}, l_{it})$ is household i's composite demand for goods consumption c_{it} and leisure and G measures the dis-utility from work. The consumption good c is a bundle of domestic and imported foreign final goods as described in Section 2.2.2.

The household's labor income is taxed at rate τ_t, such that its net labor income is

$$y_{it}^L = (1 - \tau_t) w_t h_{it} l_{it}, \tag{4}$$

where w_t is the aggregate wage rate. Given net labor income, the first-order condition for labor supply is

$$\frac{\partial G(h_{it}, l_{it})}{\partial l_{it}} = (1 - \tau_t) w_t h_{it} = \frac{y_{it}^L}{l_{it}}. \tag{5}$$

Assuming that G has a constant elasticity w.r.t. l, $\frac{\partial G(h_{it},l_{it})}{\partial l_{it}} = (1+\gamma)\frac{G(h_{it},l_{it})}{l_{it}}$ with $\gamma > 0$, we can simplify the expression for the composite consumption good, x_{it}, making use of this first-order condition (5), and substitute $G(h_{it}, l_{it})$ out of the individual planning problem:

$$x_{it} = c_{it} - G(h_{it}, l_{it}) = c_{it} - \frac{1}{1+\gamma} y_{it}^L. \tag{6}$$

When the Frisch elasticity of labor supply is constant and the tax schedule has the form (4), the dis-utility of labor is always a fraction of labor income and constant across households. Therefore, in both the household's budget constraint and felicity function, only after-tax income enters and neither hours worked nor productivity appears separately.

What remains to be determined is individual and aggregate effective labor supply. Without further loss of generality, we assume $G(h_{it}, l_{it}) = h_{it} \frac{l_{it}^{1+\gamma}}{1+\gamma}$. This functional form simplifies the household problem in the stationary equilibrium as h_{it} drops out from the first-order condition and all workers supply the same number of hours $l_{it} = L(w_t, \tau_t)$. Total effective labor input, $\int l_{it} h_{it} di$ therefore equals $L(w_t, \tau_t)$ because we normalized $\int h_{it} di = 1$.[4]

[4]This means that we can read off average productivity risk from the estimated income risk series in the

Total national firm profits are distributed in equal shares among entrepreneur households and we denote the resulting individual profit payout by Π_t^F. Similarly, total union profits are distributed in equal shares among worker households, with Π_t^U denoting the resulting individual payout. Both types of profit income are taxed at rate τ_t. What is more, households may receive *non-distortionary* targeted transfer as minimum income benefits tr_{it}. All together, after-tax non-capital income, plugging in the optimal supply of hours, is then:

$$y_{it} = \left[(1-\tau_t)w_t\right]^{\frac{1+\gamma}{\gamma}} h_{it} + (1-\tau_t)\mathbb{I}_{h_{it}\neq 0}\Pi_t^U + (1-\tau_t)\mathbb{I}_{h_{it}=0}\Pi_t^F + tr_{it}. \tag{7}$$

2.1.2 Consumption, savings, and portfolio choice

Given this after-tax non capital income, households optimize inter-temporally subject to their budget constraint:

$$c_{it} + b_{it+1} + q_t k_{it+1} = y_{it} + b_{it}\frac{R(b_{it}, R_t^b)}{\pi_t^{CPI}} + (q_t + r_t)k_{it}; \quad k_{it+1} \geq 0,\, b_{it+1} \geq \underline{B} \tag{8}$$

b_{it} is real bond holdings, k_{it} is the amount of illiquid assets, q_t is the price of these assets, r_t is their dividend, $\pi_t^{CPI} = \frac{P_t}{P_{t-1}}$ is realized domestic CPI inflation, and R is the nominal interest rate on bonds, which depends on the portfolio position of the household and the central bank's interest rate R_t^b, which is set one period before.

All households that do not participate in the capital market ($k_{it+1} = k_{it}$) still obtain dividends and can adjust their bond holdings. Depreciated capital has to be replaced for maintenance, such that the dividend, r_t, is the net return on capital. Holdings of bonds have to be above an exogenous debt limit $\underline{B}$; and holdings of capital have to be non-negative.

Substituting the expression $c_{it} = x_{it} + \frac{1}{1+\gamma}\left[(1-\tau_t)w_t\right]^{\frac{1+\gamma}{\gamma}} h_{it}$ for consumption, we obtain the budget constraint for the composite leisure-consumption good:

$$x_{it} + b_{it+1} + q_t k_{it+1} = b_{it}\frac{R(b_{it}, R_t^b)}{\pi_t^{CPI}} + (q_t + r_t)k_{it} + z_{it}, \quad k_{it+1} \geq 0,\, b_{it+1} \geq \underline{B}, \tag{9}$$

where $z_{it} = \frac{\gamma}{1+\gamma}\left[(1-\tau_t)w_t\right]^{\frac{1+\gamma}{\gamma}} h_{it} + (1-\tau_t)\mathbb{I}_{h_{it}\neq 0}\Pi_t^U + (1-\tau_t)\mathbb{I}_{h_{it}=0}\Pi_t^F + tr_{it}$ is income corrected for the dis-utility of labor.

Households make their savings choices and their portfolio choice between liquid bonds and illiquid capital in light of a capital market friction that renders capital illiquid because participation in the capital market is random and i.i.d. in the sense that only a fraction, λ, of households are selected to be able to adjust their capital holdings in a given period.

literature. Without scaling the labor dis-utility by productivity, we would need to translate productivity risk to income risk through the endogenous hour response.

The nominal interest rate on liquid bond holdings faced by households is given by:

$$R(b_{it}, R_t^b) = \begin{cases} R_t^b & \text{if } b_{it} \geq 0 \\ R_t^b + \bar{R} & \text{if } b_{it} < 0 \end{cases}. \tag{10}$$

The extra wedge for unsecured borrowing, $\bar{R}$, creates a mass of households with zero unsecured credit but with the possibility to borrow, though at a penalty rate.

Since a household's saving decision——(b'_a, k') for the case of adjustment and (b'_n, k) for non-adjustment—will be some non-linear function of that household's wealth and productivity, inflation and all other prices will be functions of the domestic joint distribution, Θ_t, of (b, k, h) in t and the foreign joint distribution, Θ_t^*. This makes Θ and Θ^* state variables of the household's planning problem and these distributions evolve as a result of the economy's reaction to aggregate shocks. For simplicity, we summarize all effects of aggregate state variables, including the distributions of wealth and income, by writing the dynamic planning problem with time-dependent continuation values.

This leaves us with three functions that characterize the household's problem: value function V^a for the case where the household adjusts its capital holdings, the function V^n for the case in which it does not adjust, and the expected continuation value, $\mathbb{W}$, over both:

$$\begin{aligned} V_t^a(b, k, h) &= \max_{k', b'_a} \{u[x(b, b'_a, k, k', h)] + \beta \mathbb{E}_t \mathbb{W}_{t+1}(b'_a, k', h')\} \\ V_t^n(b, k, h) &= \max_{b'_n} \{u[x(b, b'_n, k, k, h)] + \beta \mathbb{E}_t \mathbb{W}_{t+1}(b'_n, k, h')\} \\ \mathbb{W}_{t+1}(b', k', h') &= \lambda V_{t+1}^a(b', k', h') + (1 - \lambda) V_{t+1}^n(b', k', h'). \end{aligned} \tag{11}$$

Expectations are taken with respect to all stochastic processes conditional on current states. Maximization is subject to the corresponding budget constraint.

2.2 Firm sector

The firm sector of each country consists of five sub-sectors: (a) a labor sector composed of "unions" that differentiate raw labor and labor packers who buy differentiated labor and then sell labor services to intermediate goods producers, (b) intermediate goods producers who hire labor services and rent out capital to produce goods, (c) final goods producers who differentiate intermediate goods and then sell them to (d) goods bundlers that bundles them with foreign final goods and finally sell them as consumption goods to households and to (e) capital goods producers, who turn bundled goods into capital goods. None of these products and goods can be traded between both countries, except for the differentiated final goods.

When profit maximization decisions in the firm sector require inter-temporal decisions (i.e. in price and wage setting and in producing capital goods), we assume for tractability that they are delegated to a mass-zero group of households (managers) that are risk neutral and compensated by a share in profits. They do not participate in any asset market and have the same discount factor as all other households. Since managers are a mass-zero group in the economy, their consumption does not show up in any resource constraint and all but the unions' profits go to the entrepreneur households (whose $h = 0$). Union profits go lump-sum to worker households.

2.2.1 Labor packers and unions

Worker households sell their labor services to a unit-mass continuum of unions indexed by j. Each union differentiates the labor it purchases and sells the resulting labor variety to labor packers, which in turn supply labor services to intermediate goods producers. Labor packers produce final labor services according to the production function

$$L_t = \left(\int_0^{n_S} \hat{l}_{jt}^{\frac{\eta_W - 1}{\eta_W}} dj \right)^{\frac{\eta_W}{\eta_W - 1}} \tag{12}$$

out of labor varieties $\hat{l}_{jt}$, where $\eta_W > 1$ denotes the elasticity of substitution between differentiated labor varieties. Cost minimization by labor packers implies that each variety of labor, each union j, faces a downward-sloping demand curve

$$\hat{l}_{jt} = \left(\frac{W_{jt}}{W_t^F} \right)^{-\eta_W} L_t \tag{13}$$

where W_{jt} is the nominal wage set by union j and W_t^F is the nominal wage at which labor packers sell labor services to intermediate goods producers. Since unions have market power, they pay the households a wage lower than the price at which they sell labor to labor packers. Given the nominal wage W_t at which they buy labor from households and given the nominal wage index W_t^F, unions seek to maximize their discounted stream of profits. Throughout, we assume a zero-inflation steady state, such that all gross steady-state inflation rates equal one. However, they face a Calvo (1983) type adjustment friction with the probability λ_w of not being able to reset wages. A union that is allowed to re-optimize in period t therefore chooses W_{jt} to maximize

$$\mathbb{E}_t \sum_{s=0}^{\infty} \beta^s \lambda_w^s \frac{W_{t+s}^F}{P_{t+s}} L_{t+s} \left\{ \left(\frac{W_{jt}}{W_{t+s}^F} - \frac{W_{t+s}}{W_{t+s}^F} \right) \left(\frac{W_{jt}}{W_{t+s}^F} \right)^{-\eta_W} \right\}. \tag{14}$$

Since all unions are symmetric, we focus on a symmetric equilibrium and obtain the linearized wage Phillips curve from the corresponding first-order condition as follows,

leaving out all terms irrelevant at a first-order approximation around the stationary equilibrium:

$$\log\left(\pi_t^W\right) = \beta \mathbb{E}_t \log\left(\pi_{t+1}^W\right) + \kappa_w \left(mc_t^w - \frac{1}{\mu^W}\right), \tag{15}$$

with $\pi_t^W := \frac{W_t^F}{W_{t-1}^F} = \frac{w_t^F}{w_{t-1}^F}\pi_t^{CPI}$ being domestic wage inflation, w_t and w_t^F being the respective *real* wages for households and firms, $mc_t^w = \frac{w_t}{w_t^F}$ is the mark-down of wages the unions pay to households, W_t, relative to the wages charged to firms, W_t^F, $\kappa_w = \mu_w \frac{(1-\lambda_w)(1-\lambda_w\beta)}{\lambda_w}$, and $\mu^W = \frac{\eta_W}{\eta_W - 1}$. Total union profits accruing to workers therefore are $\Pi_t^{U,tot} = (w_t^F - w_t)L_t$.

2.2.2 Consumption Good Bundler

The consumption goods are bundles of domestically produced and imported final goods and are not traded across countries. Letting F_t denote the consumption good and S_t and N_t bundles of final goods produced in *South* and *North*, respectively, we assume the following aggregation technology

$$F_t = \left\{(1-(1-n_S)\omega_S)^{\frac{1}{\sigma}} S_t^{\frac{\sigma-1}{\sigma}} + ((1-n_S)\omega_S)^{\frac{1}{\sigma}} N_t^{\frac{\sigma-1}{\sigma}}\right\}^{\frac{\sigma}{\sigma-1}}, \tag{16}$$

$$F_t^* = \left\{(n_S\omega_N)^{\frac{1}{\sigma}} S_t^{\frac{\sigma-1}{\sigma}} + (1-n_S\omega_N)^{\frac{1}{\sigma}} N_t^{\frac{\sigma-1}{\sigma}}\right\}^{\frac{\sigma}{\sigma-1}}. \tag{17}$$

Here $\sigma > 0$ measures the terms of trade elasticity of the relative demand for domestically produced goods. $\omega_S \in [0,1]$ provides a measure for the home bias, in the sense that with $\omega_S = 1$, *South* has no home bias. The bundles of final goods produced in *South* and *North* are defined as follows:

$$S_t = \left[\left(\frac{1}{n_S}^{\frac{1}{\epsilon}} \int_0^{n_S} S_t(j)^{\frac{\epsilon-1}{\epsilon}} dj\right)\right]^{\frac{\epsilon}{\epsilon-1}}, \; N_t = \left[\left(\frac{1}{1-n_S}^{\frac{1}{\epsilon}} \int_{n_S}^{1} N_t(j)^{\frac{\epsilon-1}{\epsilon}} dj\right)\right]^{\frac{\epsilon}{\epsilon-1}}, \tag{18}$$

where $S_t(j)$ and $N_t(j)$ denote final goods produced in *South* and *North*, respectively, and ϵ measures the elasticity of substitution between final goods produced within the same country. Let $P_t(j)$ denote the price of a final good expressed in domestic currency. Then, letting $\mathcal{E}_t$ denote the nominal exchange rate (price of domestic currency in terms of foreign currency) and assuming that the law of one price holds, we have

$$P_t^*(j) = \mathcal{E}_t P_t(j), \tag{19}$$

with $\mathcal{E}_t = 1 \; \forall t$ since both countries form a monetary union.

The problem of the good bundler is to minimize expenditures subject to $F_t = C_t + I_t$, and the aggregation technologies (16) and (18). Assuming that government consumption,

G_t, is a bundle which is isomorphic to consumption goods, but consists of domestically produced goods only, global demand for a generic final good produced in *South* and *North* is given, respectively, by

$$Y_t^d(j) = \left(\frac{P_t(j)}{P_{St}}\right)^{-\epsilon} \left\{ \left(\frac{P_{St}}{P_t}\right)^{-\sigma} \left[(1-(1-n_S)\omega_S)(C_t+I_t) + (1-n_S)\omega_N Q_t^{-\sigma}(I_t^*+C_t^*)\right] + G_t \right\}, \tag{20}$$

$$Y_t^d(j)^* = \left(\frac{P_t(j)^*}{P_{Nt}^*}\right)^{-\epsilon} \left\{ \left(\frac{P_{Nt}^*}{P_t^*}\right)^{-\sigma} \left[n_S\omega_S Q_t^{\sigma}(C_t+I_t) + (1-n_S\omega_N)(I_t^*+C_t^*)\right] + G_t^* \right\}, \tag{21}$$

where the price indices are given by

$$P_{St} = \left[\frac{1}{n_S}\int_0^{n_S} P_t(j)^{1-\epsilon} dj\right]^{\frac{1}{1-\epsilon}}, \; P_{Nt} = \left[\frac{1}{1-n_S}\int_{n_S}^{1} P_t(j)^{1-\epsilon} dj\right]^{\frac{1}{1-\epsilon}} \tag{22}$$

and

$$P_t = [(1-(1-n_S)\omega_S)P_{St}^{1-\sigma} + ((1-n_S)\omega_S)P_{Nt}^{1-\sigma}]^{\frac{1}{1-\sigma}}, \tag{23}$$

$$P_t^* = [(n_S\omega_N)(P_{St}^*)^{1-\sigma} + (1-n_S\omega_N)(P_{Nt}^*)^{1-\sigma}]^{\frac{1}{1-\sigma}}. \tag{24}$$

The real exchange rate is given by

$$Q_t = \frac{P_t \mathcal{E}_t}{P_t^*}. \tag{25}$$

2.2.3 Final goods producers

Similar to unions, final goods producers in the domestic country differentiate the homogeneous domestic intermediate good and set prices. They face the global demand (20) for each good $j \in [0, n_S]$ and buy the intermediate good at the national nominal price, MC_t. As we do for unions, we assume price adjustment frictions à la Calvo (1983), with λ_Y denoting the probability of not being allowed to reset the price in a given period.

Under this assumption, a firm that is allowed to re-optimize in period t chooses the reset price p_{jt} to maximize the present value of real profits,

$$\mathbb{E}_t \sum_{s=0}^{\infty} \beta^s \lambda_Y^s \left(\frac{p_{jt}}{P_{t+s}} - \frac{MC_{t+s}}{P_{t+s}}\right) Y_{t+s}^d(j). \tag{26}$$

The corresponding first-order condition for price setting implies a domestic Phillips curve

$$\log(\pi_{St}) = \beta \mathbb{E}_t \log(\pi_{S,t+1}) + \kappa \left(\frac{mc_t}{P_{S,t}/P_t} - \frac{1}{\mu^Y}\right) \tag{27}$$

where we again dropped all terms irrelevant for a first-order approximation and have $\kappa = \mu_Y \frac{(1-\lambda_Y)(1-\lambda_Y\beta)}{\lambda_Y}$. Here, $\pi_{St} := \frac{P_{St}}{P_{St-1}}$, is gross domestic producer price inflation rate, i.e., the gross inflation rate of domestic final goods, $mc_t := \frac{MC_t}{P_t}$ are the domestic real marginal costs, and $\frac{1}{\mu^Y} = \frac{\epsilon-1}{\epsilon}$ is the inverse target markup. Total national profits accruing to domestic entrepreneurs therefore are $\Pi_t^{F,tot} = \left(\frac{P_{St}}{P_t} - mc_t\right) Y_t + \Pi_t^K$, where Π_t^K denotes profits from capital goods producers, described below.

2.2.4 Intermediate goods producers

Intermediate goods are produced with a constant returns to scale production function:

$$Y_t = Z_t (L_t)^{\alpha} (u_t K_t)^{(1-\alpha)} \tag{28}$$

where Z_t is national total factor productivity and follows an autoregressive process in logs, and $u_t K_t$ is the effective capital stock taking into account utilization, u_t, i.e., the intensity with which the existing capital stock is used. Using capital with an intensity higher than normal increases depreciation of capital according to $\delta(u_t) = \delta_0 + \delta_1(u_t - 1) + \delta_2/2(u_t - 1)^2$, which, assuming $\delta_1, \delta_2 > 0$, is an increasing and convex function of utilization. Without loss of generality, we choose δ_1 so that steady-state capital utilization is unity. Hence, δ_0 denotes the steady-state depreciation rate of capital goods.

Let mc_t be the relative price at which the intermediate good is sold to final goods producers. The intermediate goods producer maximizes profits,

$$mc_t Y_t - w_t^F L_t - [r_t + q_t \delta(u_t)] K_t, \tag{29}$$

where r_t and q_t are the rental rate of firms and the (producer) price of capital goods, respectively. The intermediate goods producer operates in perfectly competitive national markets, such that the real wage and the user costs of capital are given by the marginal product of labor and effective capital:

$$w_t^F = \alpha mc_t Z_t \left(\frac{u_t K_t}{L_t}\right)^{1-\alpha} \tag{30}$$

$$r_t + q_t \delta(u_t) = u_t (1-\alpha) mc_t Z_t \left(\frac{L_t}{u_t K_t}\right)^{\alpha} \tag{31}$$

We assume that utilization is decided by the owners of the capital goods, taking the aggregate national supply of capital services as given. The optimality condition for utilization is given by

$$q_t[\delta_1 + \delta_2(u_t - 1)] = (1-\alpha) mc_t Z_t \left(\frac{L_t}{u_t K_t}\right)^{\alpha}, \tag{32}$$

i.e., capital owners increase utilization until the marginal maintenance costs equal the marginal product of capital services.

2.2.5 Capital goods producers

Capital goods producers take the relative price of capital goods, q_t, as given in deciding about their output, i.e., they maximize

$$\mathbb{E}_0 \sum_{t=0}^{\infty} \beta^t I_t \left\{ q_t \left[1 - \frac{\phi}{2} \left(\log \frac{I_t}{I_{t-1}} \right)^2 \right] - 1 \right\}, \tag{33}$$

where $\phi > 0$ governs the size of investment adjustment costs. Optimality of the capital goods production requires (again dropping all terms irrelevant up to first order)

$$q_t \left[1 - \phi \log \frac{I_t}{I_{t-1}} \right] = 1 - \beta \mathbb{E}_t \left[q_{t+1} \phi \log \left(\frac{I_{t+1}}{I_t} \right) \right], \tag{34}$$

and each capital goods producer will adjust its production until (34) is fulfilled.

Since all capital goods producers within a country are symmetric, we obtain as the law for motion for domestic aggregate capital

$$K_{t+1} - (1 - \delta(u_t))K_t = \left[1 - \frac{\phi}{2} \left(\log \frac{I_t}{I_{t-1}} \right)^2 \right] I_t \tag{35}$$

The functional form assumption implies that investment adjustment costs are minimized and equal to 0 in steady state. Profits from capital goods production are given by $\Pi_t^K = q_t \left[K_{t+1} - (1 - \delta(u_t))K_t \right] - I_t$ and are equal to zero in steady state.

2.3 Government Sector

The two countries form a monetary union such that they run a common monetary authority. In addition, each country runs a national fiscal authority. The monetary authority controls the nominal interest rate on liquid assets in both countries, while the national fiscal authorities issue government bonds in a union-wide bond market to finance deficits, choose the tax rate in their country, and make expenditures for government consumption and their national transfer system.

2.3.1 Monetary Union

We assume that monetary policy sets the nominal interest rate, which is the same in both countries, following a Taylor (1993)-type rule with interest rate smoothing:

$$\frac{R^b_{t+1}}{\bar{R}^b} = \left(\frac{R^b_t}{\bar{R}^b}\right)^{\rho_R} \left(n_S \pi_{St} + (1-n_S)\pi_{Nt}\right)^{(1-\rho_R)\theta_\pi} \epsilon^R_t. \tag{36}$$

The coefficient $\bar{R}^b \geq 0$ determines the nominal interest rate in the steady state and π_{Nt} is the producer price inflation in *North*. The coefficient $\theta_\pi \geq 0$ governs the extent to which the central bank attempts to stabilize producer price inflation in the monetary union. $\rho_R \geq 0$ captures interest rate smoothing and ϵ^R_t is an i.i.d. monetary policy shock.

2.3.2 Fiscal Policy

The budget constraint of the national fiscal policy reads

$$\frac{P_{S,t}}{P_t} G_t + Tr_t = B_{t+1} + T_t - \frac{R^b_t}{\pi^{CPI}_t} B_t. \tag{37}$$

Hence, the government purchases G_t units of domestic goods, makes aggregate transfers Tr_t through the transfer system specified below, and services its outstanding debt, B_t. It finances these expenditures through tax revenue, T_t, and the issuance of new debt. Tax revenue is

$$T_t = \tau_t (w_t L_t + \Pi^{F,tot}_t + \Pi^{U,tot}_t). \tag{38}$$

We assume that the average tax rate is a feedback function of government debt:

$$\frac{\tau_t}{\bar{\tau}} = \left(\frac{B_{t+1}}{\bar{B}}\right)^{\gamma^\tau}, \tag{39}$$

where γ^τ governs the speed with which debt returns to its target.

2.3.3 Targeted Transfer System

The targeted transfer system provides additional resources when workers' labor income falls below a target level. For simplicity, we assume that these transfers are non-distortionary for individual labor supply by basing eligibility on labor income evaluated at aggregate hours, L_t, which households take as given. In particular, we assume the following transfer scheme:

$$tr_{it} = \mathbb{I}_{h_{it} \neq 0} \max\{0, a_1 \bar{y} - a_2 (1-\tau_t) w_t h_{it} L_t\}, \tag{40}$$

where $\bar{y}$ is the median labor income and $0 \leq a_1, a_2 \leq 1$. Because in equilibrium, all workers supply the same hours, $l_{it} = L_t$, transfers decrease with individual income with a transfer withdrawal rate of a_2 and no transfers are paid to households whose net labor income satisfies $(1-\tau_t)w_t h_{it} l_{it} \geq \frac{a_1}{a_2}\bar{y}$. Total transfer payments of the government in *South* is then

$$Tr_t = \mathbb{E}_t tr_{it}, \tag{41}$$

where again, the expectation operator is the cross-sectional average.

2.4 Goods, bonds, capital, and labor market clearing

The national labor market in *South* clears at the competitive wage given in (30). A symmetric labor market clearing condition is in place in *North*. The bond markets clear whenever the following equations hold:

$$\begin{aligned} B_{t+1} &= B^d(R_t^b, r_t, q_t, \Pi_t^F, \Pi_t^U, w_t, \pi_t^{CPI}, \tau_t, \Theta_t, \Theta_t^*, \mathbb{W}_{t+1}) - \frac{B_{Nt+1}}{Q_t} \\ &:= \mathbb{E}_t[\lambda \mathbb{B}_{a,t} + (1-\lambda)\mathbb{B}_{n,t}] - \frac{B_{Nt+1}}{Q_t}, \\ B_{t+1}^* &= B^{d,*}(R_t^b, r_t^*, q_t^*, \Pi_t^{F,*}, \Pi_t^{U,*}, w_t^*, \pi_t^{CPI,*}, \tau_t^*, \Theta_t, \Theta_t^*, \mathbb{W}_{t+1}^*) + \frac{n_S}{1-n_S} B_{Nt+1} \\ &:= \mathbb{E}_t[\lambda \mathbb{B}_{a,t}^* + (1-\lambda)\mathbb{B}_{n,t}^*] + \frac{n_S}{1-n_S} B_{Nt+1}, \end{aligned} \tag{42}$$

where $\mathbb{B}_{a,t}$, $\mathbb{B}_{n,t}$ are functions of the states (b, k, h), and depend on how the households in *South* value asset holdings in the future, $\mathbb{W}_{t+1}$, and the current set of prices (and tax rates) $(R_t^b, r_t, q_t, \Pi_t^F, \Pi_t^U, w_t, \pi_t^{CPI}, \tau_t)$.[5] Future prices do not show up because we can express the value functions such that they summarize all relevant information on the expected future price paths. Expectations in the right-hand-side expression are taken w.r.t. the distributions in both countries $\Theta_t(b, k, h)$ and $\Theta_t^*(b, k, h)$. The total net amount of foreign bond holdings in *South*, B_{Nt}, is given by the aggregation over the households' budget constraint:

$$\begin{aligned} (1-\tau_t)\left(w_t L_t + \Pi_t^{U,tot} + \Pi_t^{F,tot}\right) + \left(\frac{P_{St}}{P_t} Y_t - w_t L_t - \left(\Pi_t^{U,tot} + \Pi_t^{F,tot}\right)\right) + Tr_t + B_t R_t^b / \pi_t^{CPI} \\ + B_{Nt} R_t^b / (\pi_t^{CPI,*} Q_t) = C_t + I_t + \bar{R} BD_t + B_{t+1} + B_{Nt+1}/Q_t, \end{aligned} \tag{43}$$

where BD_t is the total amount of borrowing in *South*. Since both government bonds pay the same interest rate, we do not need to keep track of the share of domestic vs. foreign bond holdings in each household's portfolio. Equilibrium requires the total *net* amount of

[5] The same logic applies for $\mathbb{B}_{a,t}^*$, $\mathbb{B}_{n,t}^*$ in *North*.

bonds the household sectors in both countries demand to equal the supply of government bonds. In gross terms there are more liquid assets in circulation as some households borrow up to $\underline{B}$.

In addition, the national markets for capital have to clear. In *South*, we have:

$$\begin{aligned} K_{t+1} &= K^d(R_t^b, r_t, q_t, \Pi_t^F, \Pi_t^U, w_t, \pi_t^{CPI}, \tau_t, \Theta_t, \Theta_t^*, \mathbb{W}_{t+1}) \\ &:= \mathbb{E}_t[\lambda(\mathbb{K}_t) + (1-\lambda)(k)] \end{aligned} \tag{44}$$

where the first equation stems from competition in the production of capital goods, and the second equation defines the aggregate supply of funds from households in *South* - both those that trade capital, $\lambda(\mathbb{K}_t)$ and those that do not, $(1-\lambda)(k)$. Again $\mathbb{K}_t$ is a function of the current prices and continuation values. In *North*, the capital market clearing condition is symmetric.

Finally, goods markets clearing require:

$$\begin{aligned} Y_t &= (1-(1-n_S)\omega_S)\left(\frac{P_{St}}{P_t}\right)^{-\sigma}\left[C_t + I_t + BD_t\bar{R}\right] \\ &\quad + \left(\frac{P_{St}}{P_t}\right)^{-\sigma}(1-n_S)\omega_N Q_t^{-\sigma}\left[C_t^* + I_t^* + BD_t^*\bar{R}\right] + G_t \\ Y_t^* &= n_S\omega_S Q_t^{\sigma}\left(\frac{P_{Nt}^*}{P_t^*}\right)^{-\sigma}\left[C_t + I_t + BD_t\bar{R}\right] \\ &\quad + \left(\frac{P_{Nt}^*}{P_t^*}\right)^{-\sigma}(1-n_S\omega_N)\left[C_t^* + I_t^* + BD_t^*\bar{R}\right] + G_t^*. \end{aligned} \tag{45}$$

2.5 Equilibrium

A sequential equilibrium with recursive planning in our two-country model is a sequence of policy functions $\{\mathbb{X}_{at}, \mathbb{X}_{nt}, \mathbb{B}_{at}, \mathbb{B}_{nt}, \mathbb{K}_t\}$ in *South* and $\{\mathbb{X}_{at}^*, \mathbb{X}_{nt}^*, \mathbb{B}_{at}^*, \mathbb{B}_{nt}^*, \mathbb{K}_t^*\}$ in *North*, a sequence of value functions $\{V_t^a, V_t^n\}$ in *South* and $\{V_t^{a,*}, V_t^{n,*}\}$ in *North*, a sequence of prices $\{w_t, w_t^F, \Pi_t^U, \Pi_t^F, q_t, r_t, R_t^b, \pi_t^{CPI}, \pi_{St}, \pi_t^W, \frac{P_{St}}{P_t}, \tau_t, Q_t\}$ in *South* and $\{w_t^*, w_t^{F,*}, \Pi_t^{U,*}, \Pi_t^{F,*}, q_t^*, r_t^*, \pi_t^{CPI,*}, \pi_{Nt}, \pi_t^{W,*}, \frac{P_{Nt}^*}{P_t^*}, \tau_t^*\}$ in *North*, a sequence of the shock ϵ_t^R, aggregate capital, labor supply, and foreign bond holdings $\{K_t, L_t, B_{Nt}\}$ in *South* and $\{K_t^*, L_t^*\}$ in *North*, distributions Θ_t in *South* and Θ_t^* in *North* over individual asset holdings and productivity, and expectations for the distribution of future prices, Γ, such that

1. Given the functionals $\mathbb{E}_t\mathbb{W}_{t+1}$ and $\mathbb{E}_t\mathbb{W}_{t+1}^*$ for the continuation value and period-t prices, policy functions $\{\mathbb{X}_{at}, \mathbb{X}_{nt}, \mathbb{B}_{at}, \mathbb{B}_{nt}, \mathbb{K}_t\}$ and $\{\mathbb{X}_{at}^*, \mathbb{X}_{nt}^*, \mathbb{B}_{at}^*, \mathbb{B}_{nt}^*, \mathbb{K}_t^*\}$ solve the households' planning problem; and given the policy functions $\{\mathbb{X}_{at}, \mathbb{X}_{nt}, \mathbb{B}_{at}, \mathbb{B}_{nt}, \mathbb{K}_t\}$ and $\{\mathbb{X}_{at}^*, \mathbb{X}_{nt}^*, \mathbb{B}_{at}^*, \mathbb{B}_{nt}^*, \mathbb{K}_t^*\}$ and prices, the value functions $\{V_t^a, V_t^n\}$ and $\{V_t^{a,*}, V_t^{n,*}\}$ are a solution to the Bellman equation.

2. Distributions of wealth and income evolve according to households' policy functions.

3. All markets clear in every period, interest rates on bonds are set according to the central bank's Taylor rule, fiscal policies are set according to the fiscal rules, and stochastic processes evolve according to their law of motion.

4. Expectations are model consistent.

We solve the model by using the perturbation method in Bayer and Luetticke (2020).

3 Calibration

We calibrate the model to capture two salient features of the euro area: cross-country differences in public debt and in wealth inequality. We generate both from a single institutional difference across countries. Starting from *South*, which is meant to represent Italy, we introduce stronger minimum-income protection in *North* and adjust the supply of public debt so as to keep effective consumption insurance in the Euler equation—and hence the steady-state real interest rate—unchanged.[6] Stronger public insurance reduces households' demand for government bonds and, at the same time, weakens the incentives of low-wealth households to accumulate assets. As a result, *North* features both lower public debt and greater wealth inequality—a key feature of the German economy relative to Italy. We consider two calibrations for *North*. Our baseline isolates this institutional difference, with public debt adjusted to preserve the steady-state real interest rate and all remaining parameters left unchanged. In an alternative asymmetric calibration, we calibrate *North* directly to German data. In both cases, *South* is calibrated to Italy.

To pin down parameter values we match long-run averages in data or rely on values that are well established in the literature. Since our calibration strategy relies on stationary equilibrium values and a zero net-foreign asset position, the calibration is the same as for a closed economy incomplete markets model, except when trade is explicitly involved. Table 3.1 summarizes our calibration of the model parameters. We calibrate to quarterly frequency.

The labor share in production, α, is 68% corresponding to a labor income share of 62%, given a markup of 10% due to an elasticity of substitution between differentiated goods of 11. The elasticity of substitution between labor varieties is also set to 11, yielding a wage markup of 10%. The parameter δ_2 that governs the cyclicality of utilization is set such that $\frac{\delta_2}{\delta_1} = 5.0$. The investment adjustment cost is set to 4.0. We set the slopes of the price and wage Phillips curves to 0.09, which corresponds approximately to a

[6]We calibrate *South* to Italy for three reasons. It is the largest economy in the southern euro area, has persistently been among its most highly indebted members, and lacked until recently (permanent) minimum income benefits. When introducing stronger minimum income benefits, we use the German system as our benchmark. Germany is the largest economy in the northern euro area and has substantially lower public debt than Italy.

Table 3.1: Baseline Calibration—Italy

	Description	Value	Source/Target
Firms			
α	Share of labor	0.68	62% lab. income
η	Elast. of substitution	11	10% Price markup
η_W	Elast. of substitution	11	10% Wage markup
κ	Slope price PC	0.09	$\sim$ 1 year avg. price duration
κ_W	Slope wage PC	0.09	$\sim$ 1 year avg. wage duration
ϕ	Inv. adj. cost	4.0	Bayer et al. (2024a)
δ_0	Depreciation rate	0.018	Wealth Gini = 0.61
$\frac{\delta_2}{\delta_1}$	Utilization curvature	5.0	Bayer et al. (2024a)
Households			
β	Disc. factor	0.9855	B/Y = 132% annual
λ	Portfolio adj. prob.	0.038	K/Y = 330% annual
ξ	Risk aversion	4	Kaplan and Violante (2014)
γ	Inv. Frisch elast.	2	Chetty et al. (2011)
ρ_h	Pers. labor inc.	0.9815	Standard value
σ_h	STD labor inc.	0.123	Gini market incomes
ζ	Trans. prob. W to E	0.0011	T10 wealth share = 0.44
ι	Trans prob. E to W	0.0625	B50 wealth share = 0.09
$\bar{R}$	Borrowing penalty	0.019	Mass of borrowers = 0.08
Open economy			
σ	Trade-price elasticity	0.66	Standard value
ω	Home bias	0.66	Standard value
n_A	Country size	0.5	Same size
Government			
$\bar{\tau}^L$	Tax rate	0.3	Standard value
$\bar{R}^b$	Gross interest rate	1.00	zero interest-growth difference
ρ_R	Pers. in Taylor rule	0.75	standard value
θ_π	Reaction to Infl.	1.25	standard value

Notes: Parameter values that are held symmetric across countries in the baseline calibration.

Calvo non-adjustment probability of 0.75 and, equivalently, to an average price and wage duration of one year. All these parameter choices are standard values in the literature.

We calibrate the financial frictions and household preference parameters to match the wealth distribution given risk aversion and labor supply elasticities. We set relative risk aversion, ξ, to 4, following Kaplan and Violante (2014), and the Frisch elasticity, $1/\gamma$, to 0.5 following Chetty et al. (2011). The persistence of idiosyncratic income shocks, $\rho_h = 0.9815$, we set to a standard value. The standard deviation of shocks is then chosen to match income inequality in Italy; $\sigma_h = 0.123$. The remaining parameters, i.e., the discount factor, the portfolio adjustment probabilities, the borrowing penalty, the transition of workers to become entrepreneurs and the transition from entrepreneurs to become workers again, are chosen jointly to match the wealth concentration at the top, at the bottom, the share of borrowers, and the wealth Gini, as well as the total holding of government debt

and capital. To match the latter, we treat the depreciation rate as a free parameter, such that it generates the needed spread between liquid asset returns and the net marginal product of capital. The annual depreciation rate is 7.2%, reasonable for the physical assets that households hold—much of it being housing. The stationary equilibrium real rate(-growth difference) is set to a net rate of zero.

The steady state tax level is set to 0.3. We assume that monetary policy only targets inflation, as is the official mandate of the ECB, and set the Taylor coefficient to 1.25 and the smoothing parameter to 0.75. When we consider the full two-country framework, we assume that both countries are equally large and set $n_A = n_B = 0.5$. The home bias parameter, ω, and the terms of trade elasticity, σ are both set to 0.66—again standard values in the literature.

3.1 Baseline Calibration: Italy and its low-debt replica with minimum income benefits

In our baseline calibration, we construct *North* as Italy's "twin" with a lower government debt level. We microfound this difference by introducing a targeted transfer system, calibrated to the German minimum income benefits. We then adjust the government debt level in *North* such that the steady-state interest rate is the same across countries while both countries have a zero net foreign asset position.

Table 3.2 summarizes these choices and the resulting steady-state differences across countries. With a minimum income level of $a_1 = 0.5$ and a transfer withdrawal rate, $a_2 = 0.8$, which capture key features of the German minimum income benefit system (Peichl et al. 2021), we obtain a debt-to-GDP ratio of 95% in *North* compared with 132% in *South*. The level of expenditures on the minimum income benefit system is relatively small such that a government consumption share that is one percentage point of GDP lower suffices to finance these expenditures.[7]

As an additional implication of this microfoundation, minimum income benefits weaken the saving incentives of low-wealth households and therefore generate a more unequal wealth distribution (Hubbard et al. 1995). This is consistent with the Eurozone evidence documented by Pham-Dào (2016): countries with higher minimum income benefits tend to have more unequal wealth distributions.

3.2 Alternative calibration: Actual Italy and actual Germany

We also consider a version in which we calibrate *North* to directly match key statistics of Germany, see Table 3.3. The German market income inequality before transfers is somewhat higher. To match this, we set the idiosyncratic income risk $\sigma_h = 0.135$. Both the

[7]Had we calibrated to a higher real interest rate of, say, two percent, then the swap of government debt for minimum income benefits would have been completely self-financing.

Table 3.2: Baseline Calibration—Italy and its MIB replica

	Description	**South**	**North**
a_1	Transfer level	0	0.5
a_2	Transfer withdrawal rate	0	0.8
G/Y	Government consumption share	0.21	0.20
B/Y	Debt to GDP ratio	1.32	0.95
	Targets (South, Italy) / implied asset holdings (North)		
	Debt to GDP (annual)	1.32	0.95
	Capital to GDP (annual)	3.30	3.20
	Wealth Gini	0.60	0.68
	Top 10 wealth share	0.43	0.49
	Bottom 50 wealth sh.	0.10	0.05
	Mass of borrowers	0.08	0.22

German capital-to-output ratio and the German debt-to-GDP ratio are smaller. German wealth inequality is higher. We then additionally use four parameters—the discount factor, the portfolio adjustment probabilities, the transition of workers to become entrepreneurs, and the borrowing penalty—to meet these targets for Germany. This means that in our alternative calibration not only minimum income benefits but also five parameters are asymmetric across countries as summarized by Table 3.3. All other parameters remain the same as in our baseline calibration.

In order to match the data, the model requires German households to be slightly less patient, asset markets (realistically this means housing markets for most households) to be more liquid, and borrowing penalties to be higher. In addition, the mass of entrepreneurs is slightly smaller such that the individual profit payouts are somewhat larger.

4 The limiting case of a closed economy

To set the stage, we first consider both countries in isolation, setting $n_A = 1$ when analyzing *South*, and $n_B = 1$ when analyzing *North*. In this case, each country operates *de facto* as a closed economy, since the other country is small and therefore irrelevant for the outcomes in the country being studied. We will explore the full two-country case in Section 5 below.

4.1 Monetary policy transmission under different fiscal policies

We begin by analyzing the effects of a monetary policy shock. It causes output and interest rates to move in opposite directions. As a result, a contractionary monetary policy shock has unambiguously adverse fiscal consequences. We study the outcomes under two alternative fiscal policies. First, we consider a *liberal fiscal policy*, which issues additional

Table 3.3: Alternative Calibration—Asymmetric parameters

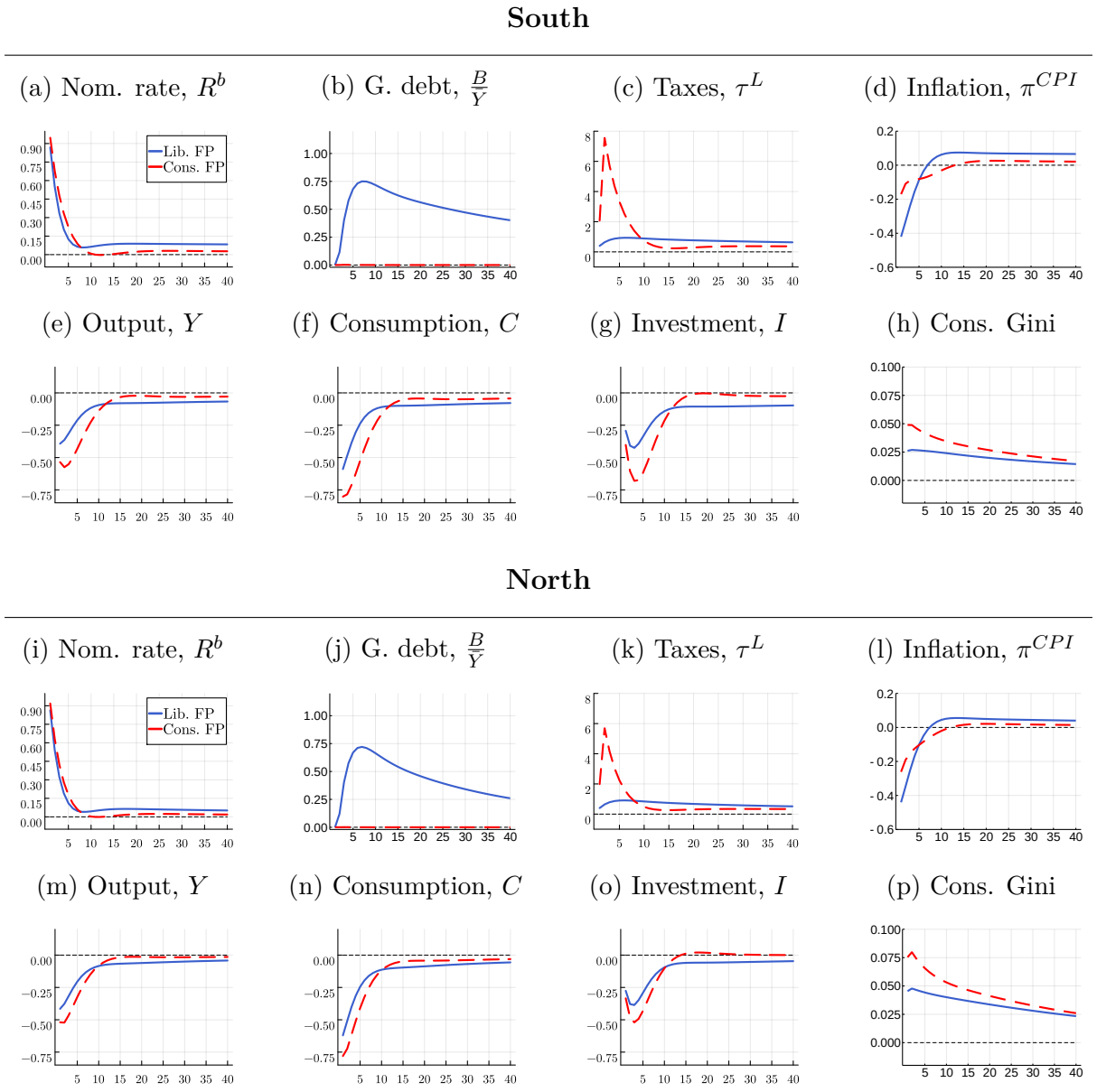
	Description	**Country A: Italy**	**Country B: Germany**
a_1	Transfer level	0	0.5
a_2	Transfer withdrawal rate	0	0.8
G/Y	Gov. cons. share	0.21	0.20
σ_h	STD labor inc.	0.123	0.135
β	Discount factor	0.9855	0.9823
λ	Portfolio adj. prob.	0.038	0.071
ζ	Trans. prob. from W to E	0.0011	0.0008
$\bar{R}$	Borrowing penalty	0.019	0.030
	Targets		
	Debt to GDP (annual)	1.32	0.71
	Capital to GDP (annual)	3.30	3.20
	Wealth Gini	0.60	0.73
	Top 10 wealth share	0.43	0.53
	Bottom 50 wealth sh.	0.10	0.02
	Mass of borrowers	0.08	0.19

government debt to buffer the fiscal implications of the shock. Formally, we implement this in the context of the monetary contraction by setting the feedback parameter γ^τ in the tax rule (39) to 0.8. As a result, the additional tax burden due to the monetary contraction is smoothed and debt is stabilized only in the long run. Second, we consider a *conservative fiscal policy* which avoids issuing additional debt in the face of a monetary contraction but relies on taxes to keep the budget balanced; formally we set $\gamma^\tau \to \infty$.

Figure 4.1 shows the impulse responses to a monetary contraction for both fiscal policies, for *South* in the upper part of the figure, and for *North* in the lower part. The solid (blue) lines represent the adjustment dynamics under the liberal fiscal policy. Interest rates are shown in panel (a). The budgetary impact of the shock is absorbed through increased debt issuance (panel b), while taxes rise only gradually (panel c). In contrast, the dashed (red) lines depict the dynamics under the conservative fiscal policy, where no additional debt is issued and taxes increase much more sharply in the short run.

The remaining panels of the figure illustrate how differences in tax policy shape the transmission of the monetary policy shock in the broader economy. First, panel (e) shows that in terms of output, the choice between liberal and conservative fiscal policy implies a trade-off between a harsher immediate contraction and a more prolonged downturn: Under conservative fiscal policy, output contracts more sharply in the short run, as both consumption (panel f) and especially investment (panel g) decline more strongly—consistent with the findings of Kaplan et al. (2018). Under liberal fiscal policy, by contrast, the recession is initially milder but considerably more persistent because taxes remain elevated for longer. As a result, output remains below its steady state

Figure 4.1: Impulse responses to a monetary policy shock



Notes: Model simulation assumes $n_A = 1$ when analyzing *South*, and $n_B = 1$ when analyzing *North*. Liberal fiscal policy ("Lib. FP") refers to $\gamma^\tau = 0.8$ and conservative fiscal policy ("Cons. FP") to $\gamma^\tau \to \infty$. *South*/*North* see Section 3. Y-axis: Percentage deviation from steady state, percentage points in the case of inflation and interest rate. X-axis: Quarters.

for an extended period. Conservative fiscal policy avoids these persistent distortions, so that output has largely recovered after about three years. Panel (d) shows that inflation falls more sharply in the short run under liberal fiscal policy than under conservative fiscal policy. The sharp increase in taxes under conservative fiscal policy induces some cost-push inflation that partly offsets the deflationary impact of the monetary tightening. Because monetary policy follows a Taylor rule, this induces a stronger nominal interest-rate response under conservative fiscal policy, thereby raising government debt-service costs over the adjustment path. This triggers a self-reinforcing tax–inflation–real-rate loop

that tends to increase the fiscal costs of conservative fiscal policy, even though the policy nevertheless entails a smaller overall output loss.[8] Finally, note that the conservative fiscal policy implies a greater increase in consumption inequality due to the monetary contraction, as the sharper short-run downturn disproportionately affects poorer and less well-insured households (panel h).

Turning to the dynamics in *North*, shown in the lower part of Figure 4.1, we observe a very similar overall pattern: the choice between liberal and conservative fiscal policy involves a trade-off between a sharper, immediate contraction and a more prolonged slump. However, there are subtle differences in the trade-offs faced by the two countries. Specifically, the required tax increase in *North* is smaller than in *South* because its lower initial level of government debt implies a smaller increase in debt-service costs for a given increase in real interest rates. As a result, the output responses under the two fiscal policies are closer to each other at each point in time. In addition, the lower initial debt level weakens the tax-inflation-real rate loop under conservative fiscal policy, because a given increase in real interest rates has a smaller effect on the debt-service costs and, thus, on the government budget. This reduces the short-run cost of conservative fiscal policy and thereby increases its cumulative output advantage.[9] In terms of inequality, however, the impact of the monetary contraction is larger under both fiscal policies compared to what is observed in *South*.

4.2 Who votes fiscally conservative?

Fiscal policy shapes the transmission of monetary policy and gives rise to macroeconomic trade-offs. How individual households assess these trade-offs, however, varies systematically with their own economic circumstances. To establish this formally, we compute the one-sided welfare impact of each policy for every individual household, where households are defined by their position in the joint distribution of liquid and illiquid asset holdings and income. We then compare household-level welfare under liberal and conservative fiscal policy.[10]

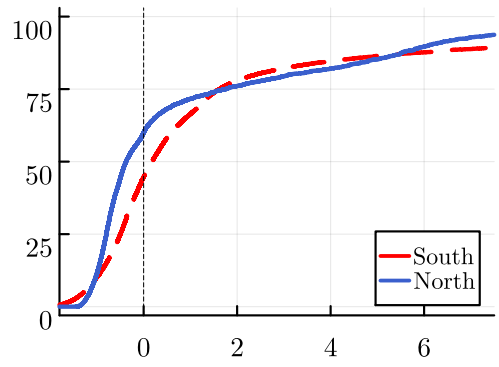
[8] Specifically, the discounted cumulative output loss is 12% larger under liberal than under conservative fiscal policy.

[9] In *North* the discounted cumulative output losses are 18% larger under liberal fiscal policy than under conservative fiscal policy. The trade-off which governs the cumulative output loss is not specific to our quantitative HANK model. In a representative-agent New Keynesian model with distortionary taxation, fiscal front-loading lowers cumulative output losses for a given cumulative tax adjustment because future tax hikes depress activity already in anticipation. Yet it can raise the required adjustment through the tax–inflation–real rate loop, especially when initial government debt is high. Appendix E zooms in on this trade-off in a parsimonious three-period model and shows that higher steady-state debt can weaken and eventually overturn the advantage of front-loading the fiscal adjustment.

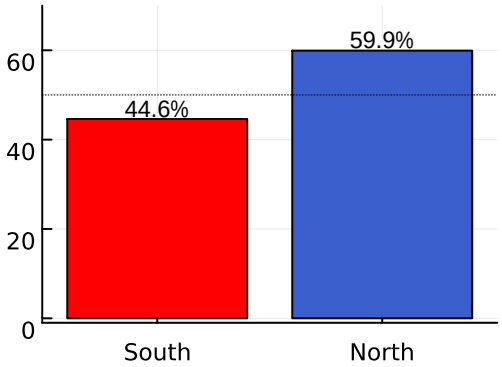
[10] We express welfare effects as consumption-equivalent variations, that is, as the permanent proportional change in the composite consumption good, x, that generates the same first-order change in household welfare. The comparison is conditional on the realized aggregate shock and the household's current asset and productivity state, with welfare averaged over the current portfolio-adjustment opportunity. We abstract from questions of optimal policy design or from how systematic policy responses might shape

Figure 4.2: Voting for conservative fiscal policy

(a) Preference distribution: lib. vs. cons.



(b) Support for cons. fiscal policy



Notes: The left panel shows the cumulative distribution function of the welfare differences between liberal ("lib.") and conservative fiscal policy ("cons. FP"): positive (negative) values on the horizontal mean that welfare is higher under the liberal (conservative) policy. *South*/*North* see Section 3. The right panel displays the share of votes in favor of conservative fiscal policy in each country. Y-axes are in percent.

The left panel of Figure 4.2 shows how these welfare differences are distributed in both countries. It plots the cumulative distribution of households' preferences, defined as the welfare difference between liberal and conservative fiscal policy. Negative values on the horizontal axis indicate a preference for conservative fiscal policy, while positive values indicate a preference for liberal fiscal policy, with larger absolute values indicating larger welfare differences between the two fiscal policies. The cumulative distribution has two distinct features: First, in both countries, fairly large segments of the population lie on either side of the indifference point (marked by the dashed line at zero). Accordingly, under both policies a large part of the population is relatively better off than under the alternative. Second, welfare differences are highly asymmetric: to the far right of the indifference point, a non-negligible group of households experiences large gains from liberal fiscal policy, whereas comparably large gains from conservative fiscal policy are much less common. Figure A.8 in the Appendix shows that these large gains from liberal fiscal policy arise primarily for households in low-wealth states. Intuitively, these households are less able to absorb the deeper immediate recession under conservative fiscal policy and therefore strongly value the smoother adjustment under liberal fiscal policy. As a result, we find that liberal fiscal policy yields higher *average* welfare in both countries.

We then assume—in line with the fundamental premise underlying our paper—that these differences in welfare effects determine households' attitudes toward, and support for or opposition to, fiscal policies: Households "vote" for the fiscal policy that makes them better off in terms of welfare. The bars in the right panel show the voting shares in favor of conservative fiscal policy in both countries. The key take-away is that the

welfare outcomes, as in Gornemann et al. (2021).

majority support differs across both countries: In *South* 44% prefer conservative fiscal policy, implying that the "median voter" supports liberal fiscal policy. In *North*, by contrast, 60% of households prefer conservative fiscal policy. Thus, the median voter in *North* supports conservative fiscal policy even though it delivers lower *average welfare*—a result familiar from many political-economy contexts.

Mutatis mutandis, these results also imply that following a monetary *expansion*, households in *South* would prefer a balanced-budget policy (that is, $\gamma^\tau \to \infty$), which—in the case of a monetary expansion—amounts to a liberal fiscal policy: it prevents any fiscal surplus that would otherwise contribute to reducing the level of debt.[11] Instead, the median voter in *North* would indeed opt for running a fiscal surplus (i.e., $\gamma^\tau = 0.8$). Hence, in this sense, *North* prefers a more conservative fiscal policy relative to *South* also after a monetary expansion. In sum, if fiscal policy were subject to majority voting, *North* would prefer a more conservative fiscal policy regardless of whether there is a monetary contraction or a monetary expansion, whereas *South* would vote in favor of a more liberal fiscal policy.

The top panels of Figure 4.3 systematically link voting behavior to a household's position in the income and wealth distributions in both countries. Two insights emerge. First, and most importantly, support for conservative fiscal policy is generally higher in *North* than in *South* along both dimensions. Second, there are some noticeable cross-country differences in how policy preferences change in income and wealth. Panel (a) shows that support rises almost monotonically with income in *North*, with the highest income decile being the only exception. In *South*, by contrast, support is much more non-monotonic: it rises up to the upper-middle part of the income distribution, but falls sharply among the highest-income households. Panel (b) reveals a similar cross-country difference along the wealth distribution. Support increases sharply over the lower wealth deciles and remains substantially higher in *North*, especially among wealthier households.

To see what gives rise to these patterns, we sequentially decompose the difference in support for conservative fiscal policy between South and North into differences in private insurance, government insurance, and macroeconomic adjustment. First, we recompute the voting shares for conservative fiscal policy in *South*, replacing, all else equal, the *South* wealth distribution with that of *North*—the distribution that would have emerged if *South* had implemented the same MIB transfer scheme as *North* and which features a lower level of "private insurance". The orange bars in panel (c) show that, as a result, support for conservative fiscal policy declines across the entire income distribution.

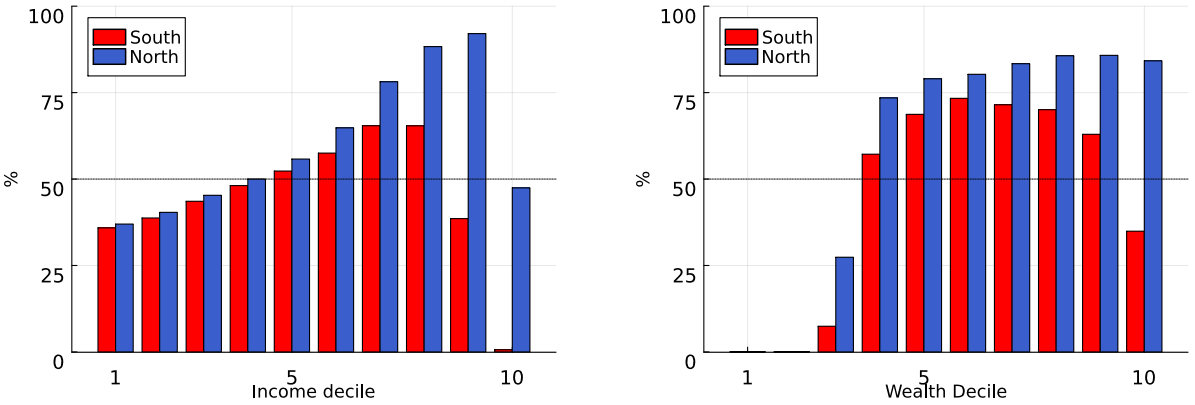
Second, starting from this counterfactual distribution, we additionally assume that households receive the income benefits provided under the MIB scheme operating in *North*, while maintaining the general-equilibrium adjustment paths of *South*. The blue bars in

[11] Given that the model solution is linear in aggregate variables, in response to a monetary expansion, 55.4% (44.6%) of households in *South* would be better (worse) off under a balanced-budget fiscal policy.

Figure 4.3: Support for conservative fiscal policy

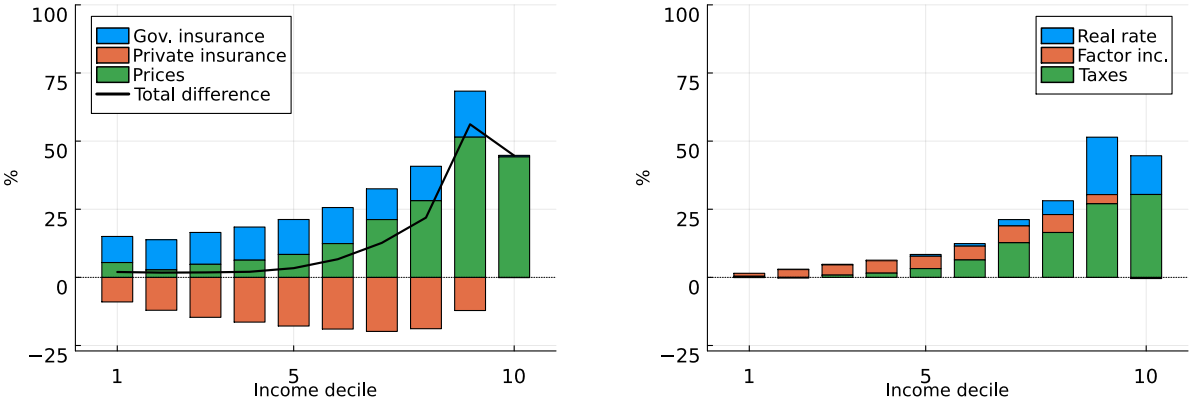
(a) Votes along income distribution

(b) Votes along the wealth distribution



(c) Overall decomposition

(d) Zooming in on the GE prices



Notes: The top panels show the share of households in favor of conservative fiscal policy within each income (left) and wealth (right) decile. Panel (c) decomposes the differences in support for conservative fiscal policy within each income decile between *South* and *North*. See main text for a description of how we compute these decompositions. Panel (d) further decomposes the price channel ("prices"). *South*/*North* see Section 3. Y-axis denoting percentages.

panel (c) of Figure 4.3 illustrate how such "government insurance" shifts voting behavior: across the entire income distribution, support for conservative fiscal policy increases: better-insured households are generally more willing to tolerate the deeper immediate recession associated with conservative fiscal policy.

Quantitatively, the effects of government insurance and private insurance on voting behavior almost exactly offset each other.[12] What remains is the difference in macroeconomic adjustment between the two countries. Equivalently, this remaining component corresponds to additionally replacing the equilibrium price responses of *South* with those prevailing in *North*, after the wealth distribution and MIB scheme have already been replaced. This macroeconomic component accounts for most of the difference in support between the two countries: The green bars in panel (c) show that support for conservative

[12]This suggests that the introduction of MIBs can improve households' willingness to support short-run contractionary policies, but this additional support is offset once the wealth distribution has settled at its new equilibrium.

fiscal policy increases across the entire income distribution in the *South* once households are confronted with the counterfactual equilibrium prices of the *North* ("Prices"). This is intuitive: as shown above the macroeconomic impact of the conservative fiscal policy is relatively better in *North* because of its more favorable initial fiscal position. Panel (d) further decomposes the price responses in Panel (c) into changes in factor income (consisting of capital, labor, and profit incomes), tax adjustments, and the real return on liquid assets. All three components positively contribute to the increased support for conservative fiscal policy, albeit with varying strengths across the income distribution. Differences in factor income dynamics increase support particularly among lower- and middle-income households. Differences in tax adjustments matter most for middle- and high-income households.[13] Finally, differences in real-return dynamics matter mainly for high-income households.[14]

4.3 Funding higher government spending: Proof of concept

So far, we have focused on the fiscal policy response to a monetary policy shock. The nature of the shock also matters for policy preferences, as we now show by examining fiscal attitudes when the economy is subject to an exogenous increase in government spending. This scenario also allows us to relate our results to a 2025 survey of German households on the country's shift toward a more expansionary fiscal policy to finance higher government expenditure. We show that heterogeneity in the welfare effects across households helps explain the variation in survey responses. This exercise serves as a proof of concept: differences in the economic effects of alternative policies, as predicted by our model, can help rationalize voter preferences revealed in the survey data.

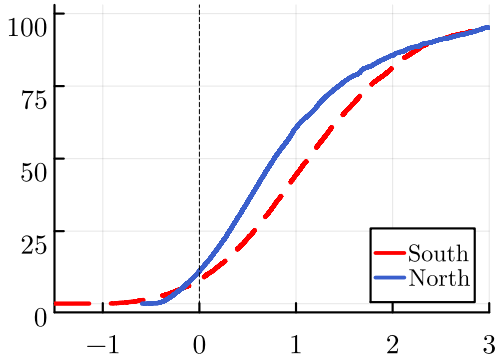
Specifically, we consider an exogenous shock to government spending, G_t, which follows an AR(1) process with a persistence parameter of 0.85. Liberal and conservative fiscal policies are given, as before, by $\gamma^\tau = 0.8$, such that government spending is temporarily debt-financed (liberal policy), and by $\gamma^\tau \to \infty$, such that the budget remains balanced (conservative policy). Figure 4.4 plots the cumulative preference distributions in *North* and *South* in panel a), and maps these into voting shares for conservative fiscal policy, shown in panel b). Qualitatively, we observe the same pattern as in the case of the monetary contraction analyzed above: support for conservative fiscal policy is higher in *North*. In absolute terms, however, support remains low in both countries—only 8.4% in *South* and

[13] Intuitively, the overall tax path is more favorable to conservative fiscal policy under the *North* price responses due to the weaker tax-inflation-real rate loop. Second, high-income households are particularly sensitive to the large contemporaneous tax increases required under conservative fiscal policy in *South*, because they expect their incomes to decline in the future.

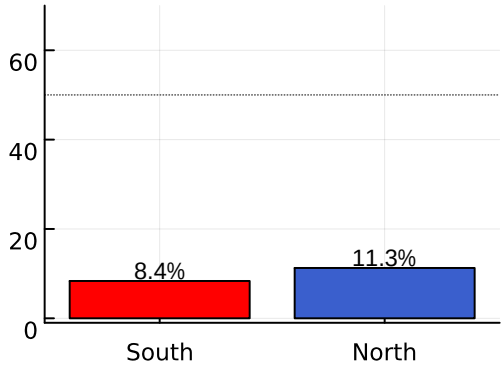
[14] Under the *South* price responses, liberal fiscal policy generates a more pronounced long-run increase in real returns than in *North*. This benefits high-income households that build up liquid assets because they expect their future incomes to decline, making liberal fiscal policy relatively more attractive for them in *South*.

Figure 4.4: Voting for conservative fiscal policy: Government spending

(a) Preference distribution: lib. vs. cons.



(b) Support for cons. fiscal policy



Notes: The left panel shows the cumulative distribution function of the welfare differences between liberal ("lib.") and conservative fiscal policy ("cons. FP"): positive (negative) values on the horizontal mean that welfare is higher under the liberal (conservative) policy. *South*/*North* see Section 3. The right panel displays the share of votes in favor of conservative fiscal policy in each country. Y-axes are in percent.

11.3% in *North*. Most strikingly, even in *North*—where a majority supported conservative fiscal policy in response to monetary contractions—there is now an overwhelming majority in favor of liberal fiscal policy.

Two forces drive this result. First, higher government spending is inflationary and prompts the central bank to raise interest rates. In response to this monetary tightening, voters in *North* become more willing to accept conservative fiscal measures, as shown above, thereby explaining the relative difference in support across countries. Second, conservative fiscal policy requires immediate tax increases not only to offset higher debt-service costs but also to finance the increase in government spending itself, regardless of a country's initial debt level. Government spending shocks therefore generate a much stronger tax–inflation–real-interest-rate feedback loop than a pure monetary tightening, making conservative fiscal policy substantially more costly in both countries. As a result, its appeal declines in both countries.

Still, within countries, the welfare costs of different fiscal policies vary substantially across households. We illustrate this for *North* in Figure A.9 in the Appendix. These within-country differences provide an opportunity to confront our theory with survey evidence for Germany. In the spring of 2025, the funding of additional government spending on public infrastructure became a particularly salient issue, as the incoming coalition government managed to relax the constitutional deficit limit, the so-called "debt brake," to permit higher borrowing. This effectively shifted Germany's fiscal policy response from conservative to liberal in the face of increased government spending.

Against this background, we ran a survey in May 2025 to elicit respondents' attitudes toward easing the constitutional limit on debt financing.[15] Concretely, survey participants

[15]For details on the survey as well as the complete set of questions, see Appendix D.

Table 4.4: Political support of liberal debt policy to finance government spending, survey

	linear	ordered probit	ordered logit
log CEV	0.18	0.16	0.28
	(0.09)	(0.08)	(0.15)
N	452	452	452

Notes: The table shows the regression coefficients of a regression that has survey answers to the question of whether households view the exemption of planned infrastructure investments from the German debt brake positively on the left hand side (five-point Likert scale, from strong disagreement to strong agreement). The answers are regressed on the imputed consumption-equivalent variation. Survey respondents are categorized based on the income and wealth deciles they belong to. the regression only uses those respondents that view the German debt brake positively in general (Answers 4 or 5 on the Likert scale). Regressions control for household size (OECD equivalence scale), age, and general political leaning. Standard errors in brackets are clustered at the income-wealth decile combination.

were asked whether they approve of a constitutional reform which allows the government to fund a significant share of infrastructure spending by issuing new debt.[16] The survey also asked about respondents' income and total wealth such that we can locate them in the income and wealth distribution.

For each combination of income and wealth deciles, we calculate the average difference between liberal and conservative fiscal policies expressed in consumption-equivalent variation (CEV) in response to the government spending shock, see again Figure A.9. We then estimate the response x_j of respondent j as a function of the CEV of their income and wealth decile, $i(j)$ and $w(j)$, respectively:

$$x_j = \alpha + \beta \log CEV(i(j), w(j)) + \gamma \text{ control}_j, \tag{46}$$

where responses, x_j, can be one of the following: "strongly disagree," "tend to disagree," "neither agree nor disagree," "tend to agree," or "completely agree" with the proposal to exempt the spending increase from the deficit limit.[17]

Survey participants were also asked about their general views on the debt brake. Because support for a specific spending exemption is less informative among respondents who oppose the debt brake in principle, we restrict our analysis to those who viewed the introduction of Germany's debt brake in 2008 favorably. We account for household size using the OECD equivalence scale, as only unadjusted income and wealth deciles are available. We also control for respondents' broader political orientation and include age linearly.

[16] Because we conducted the survey around the time such an exception was added to the German constitution, survey participants were likely to have the concrete reform in mind when answering the question.

[17] For all income-wealth groups j CEV is positive for the spending exercise, such that the log exists.

Table 4.4 shows the results for alternative specifications: a linear regression, an ordered probit, and an ordered logit. The result is robust: households for which the model predicts a higher material benefit from financing the planned fiscal expansion by debt issuance (liberal fiscal policy) are more likely to support the constitutional reform that allows for exactly this debt issuance. Although the number of observations is limited and the actual German reform might be linked to further policy connotations, we consider this regression result to be a clear proof of concept that goes beyond our motivating anecdotal evidence.

5 A Heterogeneous Monetary Union

We now extend our analysis to the full two-country framework, setting $n_A = n_B = 0.5$. We first show that our main results remain largely unchanged when the two countries interact within a monetary union and vote on union-wide fiscal policies. We then use the $HANK^2$ setup to explore the cross-border implications of the union's political economy. In particular, we show that in each member state, a large majority prefers the other country to adopt conservative fiscal policies.

5.1 Union-wide fiscal rules

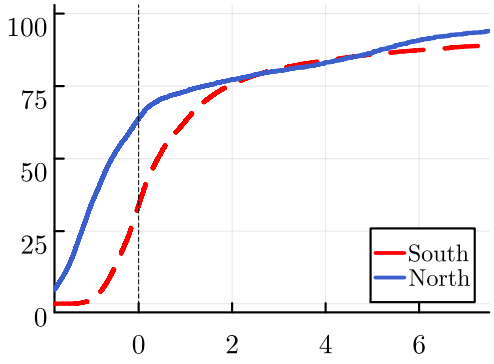
We start by exploring the case, in which both countries adopt the same fiscal policy in the face of a monetary contraction—either liberal ($\gamma^\tau = 0.8$) or conservative ($\gamma^\tau \to \infty$). The aggregate impulse responses to a common contractionary monetary policy shock—reported in the Appendix to economize on space (Figure B.10)—closely mirror our closed-economy results in Section 4. This is because, despite sizable differences in the degree of household heterogeneity, these country differences have only a minor impact on the dynamics triggered by a common shock (Bayer et al. 2024b).

Consistent with these similar aggregate dynamics, the within-country voting patterns also closely resemble our earlier findings. Figure 5.5 shows the cumulative preference distributions in *South* and *North* in panel (a) and how they map into voting shares for conservative fiscal policy in panel (b). As before, the median voter in *North* prefers conservative fiscal policy, while the median voter in *South* prefers liberal fiscal policy.

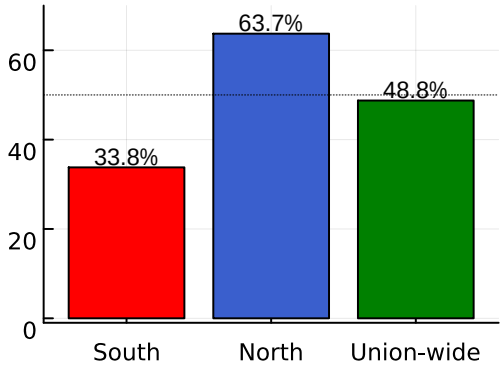
Quantitatively, the divergence in fiscal attitudes across countries is somewhat larger in the monetary union than in the closed economy case studied above: in *South*, only 33% instead of 45% vote for conservative fiscal policy, while in *North*, 64% instead of 60% do so. Relative to the closed-economy benchmark, two opposing macroeconomic channels affect the relative appeal of the fiscal policies. First, the common monetary policy spreads part of the tax–inflation–interest-rate feedback generated by front-loaded fiscal adjustment across countries. Because the common monetary authority responds to union-wide rather than national inflation, the fiscal burden generated by this feedback is

Figure 5.5: Voting for conservative fiscal policy in the monetary union

(a) Preference distribution: lib. vs. cons.



(b) Support for cons. FP



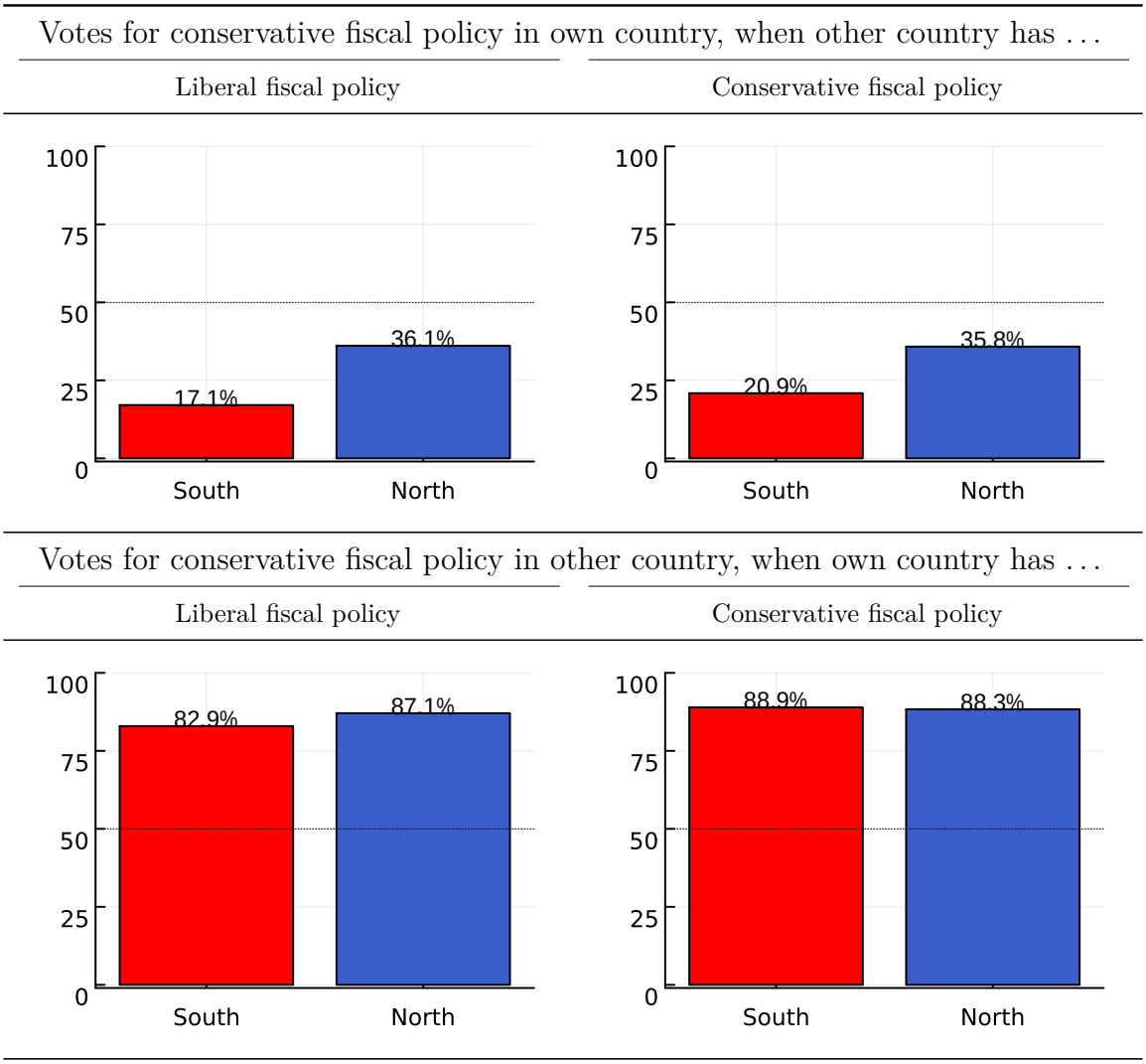
Notes: The left panel shows the cumulative distribution function of the welfare differences between liberal ("lib.") and conservative fiscal policy ("cons. FP"): positive (negative) values on the horizontal mean that welfare is higher under the liberal (conservative) policy. *South*/*North* see Section 3. The right panel displays the share of votes in favor of conservative fiscal policy in each country. Y-axes are in percent.

smaller for high-debt *South* than in the closed-economy benchmark and larger for low-debt *North*. Conservative fiscal policy therefore becomes relatively less costly in *South* and more costly in *North*, which, ceteris paribus, leads to greater convergence in fiscal attitudes across countries.

Second, because countries specialize in the production of different goods, fiscal adjustment can shift part of its burden abroad through the terms of trade. With nominal rigidities, the timing of tax adjustments matters: in the short run, nominal rigidities mitigate the impact of higher taxes on prices, but less so over time. As a result, front-loaded tax increases fall mainly on domestic residents. In contrast, deferring taxes allows a greater share of the future tax burden to be shifted onto foreigners through higher domestic prices. Since *South* faces the larger fiscal adjustment, delaying taxes generates stronger upward pressure on its domestic prices relative to those in *North*. The resulting terms-of-trade adjustment therefore allows *South* to shift more of its fiscal burden abroad. Liberal fiscal policy therefore becomes relatively more attractive in *South*, while conservative fiscal policy becomes relatively more attractive in *North*, which, ceteris paribus, leads to greater divergence in fiscal attitudes across countries. Our results suggest that this second channel dominates, so that the monetary union ultimately increases divergence in fiscal attitudes. Lastly, at the union level, a slim majority favors liberal fiscal policy.

As a robustness exercise, we recalibrate *North* to reflect "actual Germany" instead of "Italy with MIB" (see Section 3). Our results remain qualitatively unchanged. Quantitatively, the difference between liberal and conservative fiscal policy in *North* becomes somewhat smaller, reflecting the lower initial debt level and hence the smaller required tax adjustment under the alternative calibration (see Figure B.11 in the Appendix). Moreover, although a majority in *North* still favors conservative fiscal policy, support falls from

Figure 5.6: Support for conservative fiscal policy with country-specific rules



Notes: Displayed is the share of households in favor for conservative fiscal policy. The top panels show voting outcomes when residents decide on their own country's fiscal response; the bottom panels show voting outcomes when they decide on the foreign country's fiscal response. *South*/*North* see Section 3. Y-axis denoting percentages.

64% to 57% (see Figure B.12 in the Appendix). While the lower debt level of "actual Germany" tends to increase support for conservative fiscal policy, this effect is more than offset because matching the German economy requires a lower discount factor in the *North* calibration, which shifts the evaluation of the intertemporal trade-off towards avoiding a deeper recession today.

5.2 Country-specific fiscal rules

We now turn from a common fiscal response to country-specific policies, reflecting the fact that fiscal decisions in the euro area are made primarily at the national level. Specifically, we once more calculate the shares of households in favor of a given policy rule in each country, this time conditional on a given fiscal policy adopted by the other country.

Figure 5.6 displays the results, conditional on a monetary contraction. Two observations are noteworthy. First, neither country prefers to unilaterally adopt conservative fiscal policy. Even *North*, which prefers union-wide fiscal conservatism, would not choose to deviate unilaterally. Second, both countries strongly prefer the other country to adopt a conservative fiscal policy. These patterns follow naturally from the terms-of-trade channel discussed above: with nominal rigidities, delaying fiscal adjustment allows a larger share of the tax burden to be shifted abroad. Households therefore favor a more gradual adjustment at home, while preferring the partner country to front-load its adjustment.

6 Conclusion

Who is fiscally conservative? Who is fiscally liberal? Rather than attributing fiscal attitudes to deep-rooted national preferences, we seek to answer these questions based on a person's economic circumstances—both idiosyncratic and aggregate. While we do not rule out that national preferences may matter at times, our analysis explores how far differences in fiscal attitudes can be rationalized by economic circumstances alone. To this end, we use the household heterogeneity embedded in HANK to study the political support for alternative national fiscal stances within a monetary union.

Specifically, we develop a two-country version of the model, HANK^2, and calibrate it to capture key features of the EA, in particular its *South* and *North*. A central aspect of the calibration is that government debt is higher and wealth inequality lower in *South*. Conditional on a monetary tightening, we find majority support for conservative fiscal policy in *North* and for liberal fiscal policy in *South*, broadly in line with prevailing views on fiscal attitudes in Europe. In our model, however, this difference reflects economic conditions rather than national identities. Crucially, it is driven primarily by the different macroeconomic costs of fiscal adjustment associated with different initial debt levels. Moreover, political majorities are not fixed: they vary with the nature of the underlying shock. Following an increase in government spending, for instance, the median voter in *North* also prefers liberal fiscal policy—by a wide margin.

Our analysis is relevant for policy because it highlights a dimension of heterogeneity within currency unions that has received little attention: differences in economic conditions across countries can critically shape popular support for fiscal policy. Harmonizing these conditions may therefore be important for the political viability of currency unions. We leave a formal assessment of this conjecture to future research.

At a conceptual level, our analysis reveals the political-economy dimension inherent in modern macroeconomic models with rich household heterogeneity. This perspective extends naturally beyond monetary unions and can be applied to a broad range of questions in monetary and fiscal policy.

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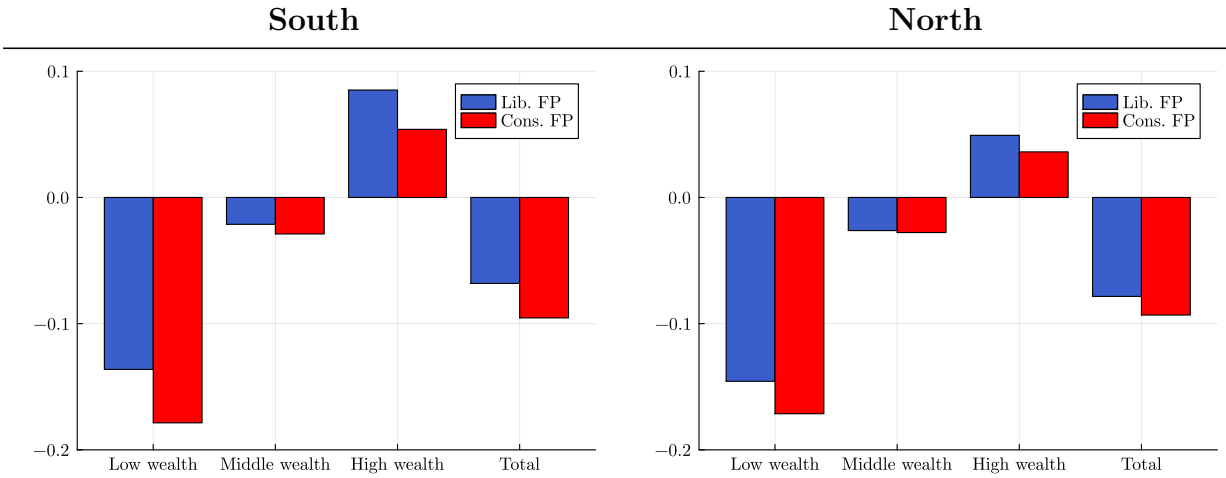
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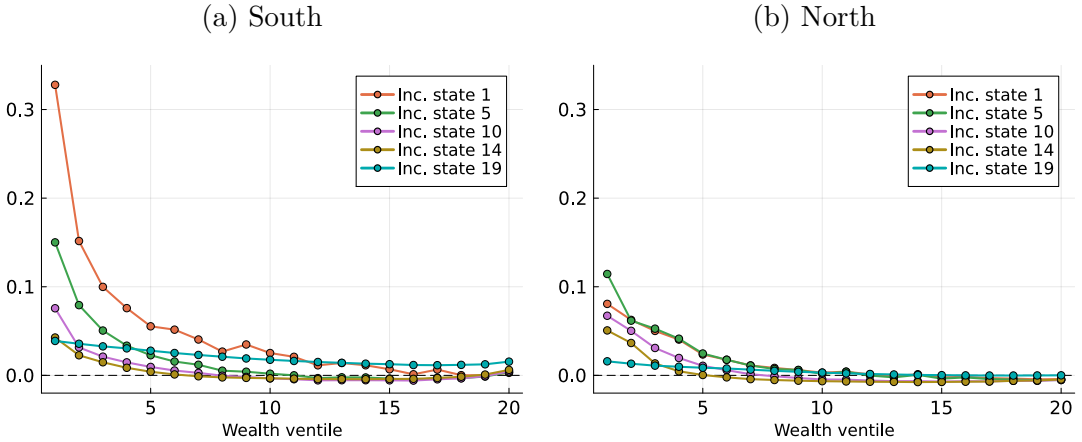
A Further Results Closed-economy

Figure A.7: Welfare impact of monetary contraction



Notes: Average willingness to pay, expressed in terms of consumption equivalence, by wealth group after a contractionary monetary policy shock. "Low" refers to the bottom 50%, "Middle" to the 50–90% percentile group, and "High" to the top 10%. Wealth denotes total wealth, including liquid and illiquid assets. Liberal fiscal policy ("Lib. FP") corresponds to $\gamma^\tau = 0.8$, and conservative fiscal policy ("Cons. FP") to $\gamma^\tau \to \infty$. *South*/*North* see Section 3.

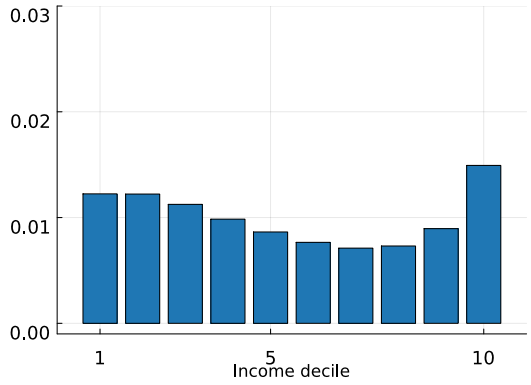
Figure A.8: Welfare gains from liberal fiscal policy after a monetary tightening



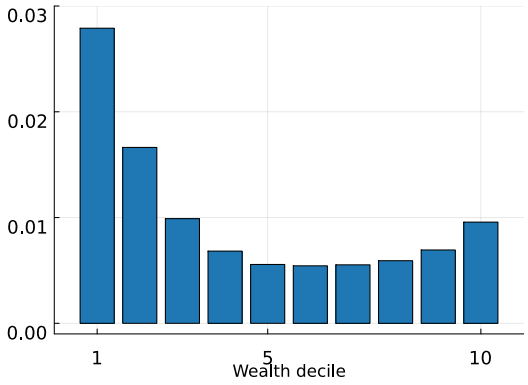
Notes: Average welfare differences between liberal and conservative fiscal policy after a monetary tightening, by wealth (bonds + capital) ventile and income state. Positive values indicate that welfare is higher under liberal fiscal policy, while negative values indicate that welfare is higher under conservative fiscal policy. The figure shows the welfare effect for households located at a given point in the state space; it does not indicate how many households are located in each state.

Figure A.9: Welfare differences across income and wealth deciles between liberal and conservative fiscal policy in *North* after a government spending shock

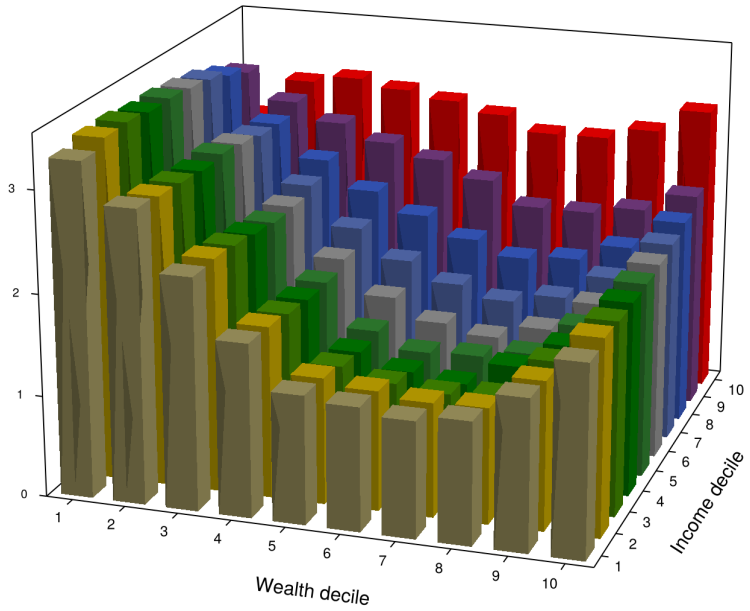
(a) Income



(b) Wealth



(c) Joint income-wealth, $log(CEV * 1000)$



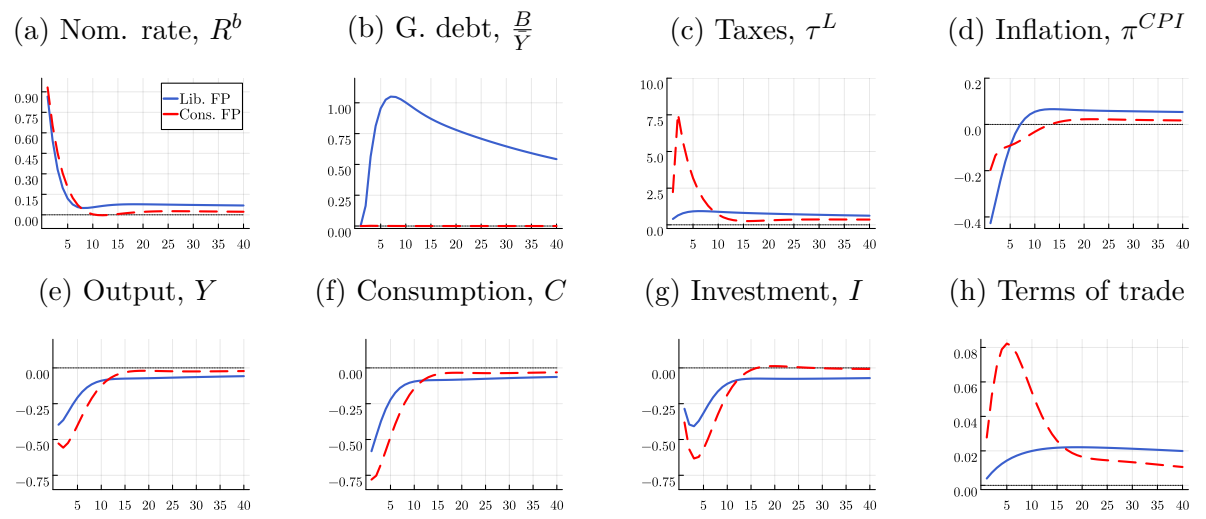
Notes: Average welfare differences between liberal ("lib.") and conservative fiscal policy ("cons. FP") after a government spending shock: positive (negative) values mean that welfare is higher under the liberal (conservative) policy. Panel (a) groups households by income deciles (b) by wealth deciles and (c) by joint income-wealth deciles. Welfare expressed in average willingness to pay in terms of consumption equivalence (CEVs).

B Further Results Two-country

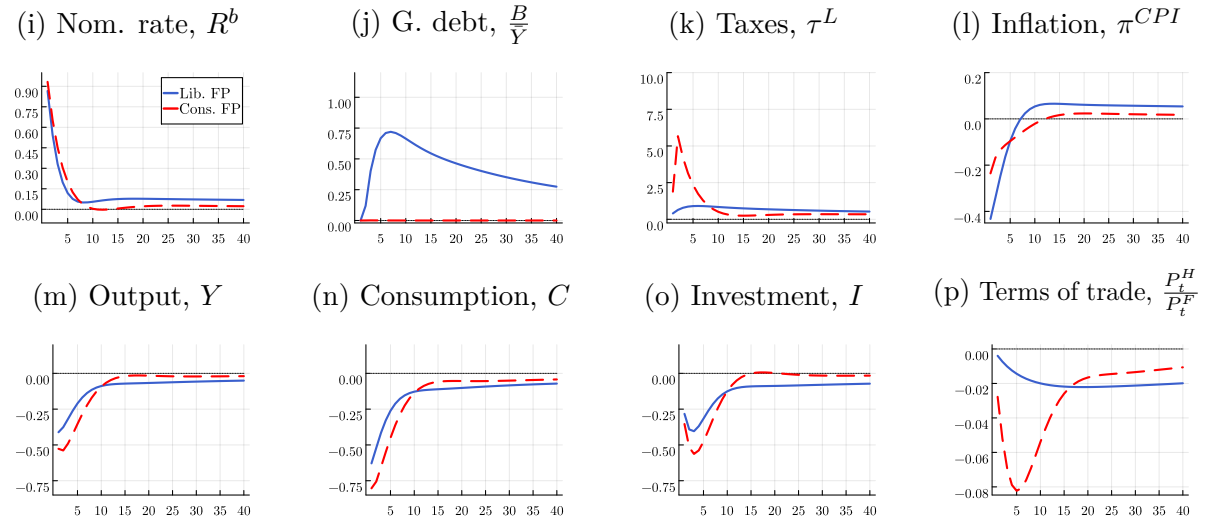
B.1 IRFs after monetary policy shock in monetary union

Figure B.10: Impulse responses to a monetary policy shock in the monetary union

South



North



Notes: The figure shows impulse responses to a common monetary policy contraction in the monetary union. Liberal fiscal policy ("Lib. FP") refers to $\gamma^\tau = 0.8$ and conservative fiscal policy ("Cons. FP") to $\gamma^\tau \to \infty$. *South*/*North* see Section 3. Y-axis: Percentage deviation from steady state, percentage points in the case of inflation and the interest rate. X-axis: Quarters.

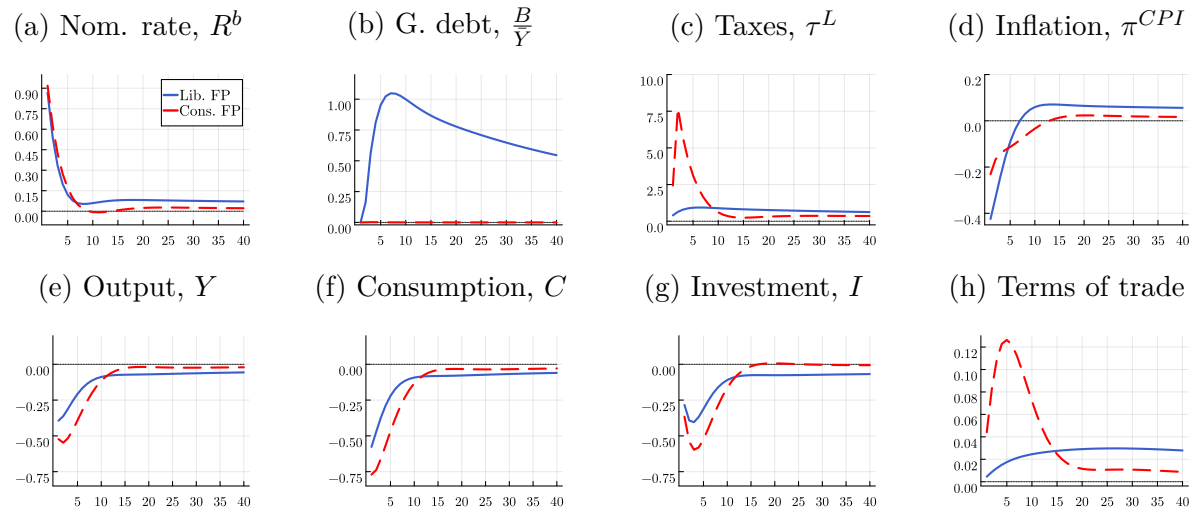
B.2 Alternative Calibration: Italy and actual Germany

Figure B.11 shows that the lower debt level of actual Germany reduces the required tax adjustment in *North* and thereby narrows the difference between liberal and conservative

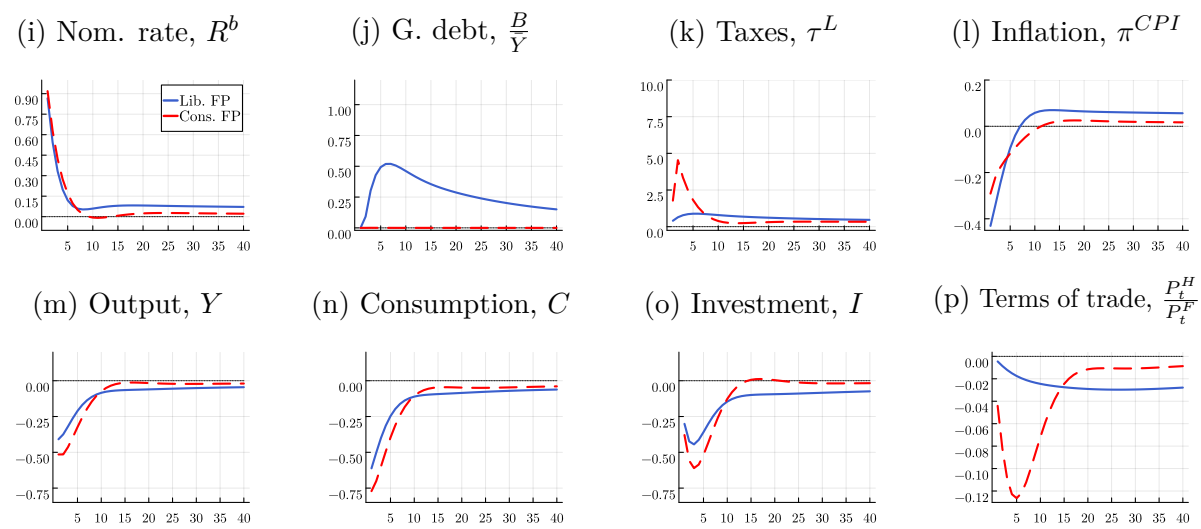
fiscal policy. Overall, however, the qualitative findings remain very similar to our baseline.

Figure B.11: Impulse responses to a monetary policy shock: Alternative calibration

South



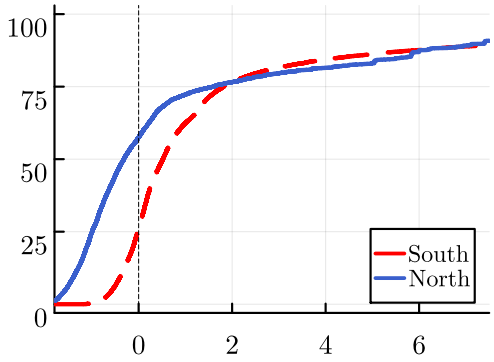
North (actual Germany)



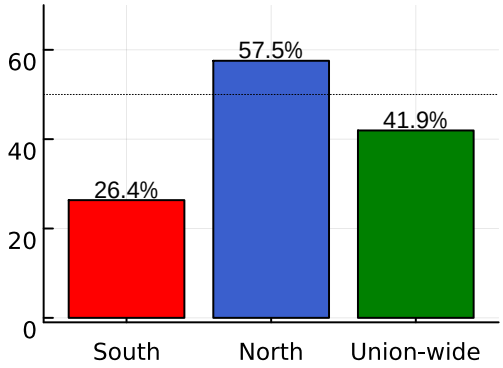
Notes: The figure shows impulse responses to a common monetary policy contraction under the alternative calibration in which *North* is calibrated to Germany, while *South* remains as in the baseline. Liberal fiscal policy ("Lib. FP") refers to $\gamma^\tau = 0.8$ and conservative fiscal policy ("Cons. FP") to $\gamma^\tau \to \infty$. See Section 3 for details on the calibration. Y-axis: Percentage deviation from steady state, percentage points in the case of inflation and the interest rate. X-axis: Quarters.

Figure B.12: Voting for conservative fiscal policy in the monetary union: Alternative calibration

(a) Preference distribution: lib. vs. cons.



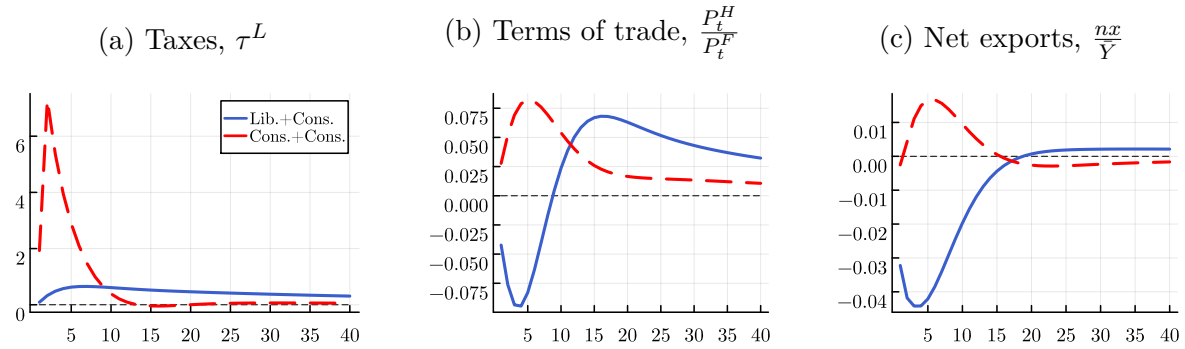
(b) Support for cons. FP



Notes: The left panel shows the cumulative distribution function of the welfare differences between liberal ("lib.") and conservative fiscal policy ("cons. FP"): positive (negative) values on the horizontal mean that welfare is higher under the liberal (conservative) policy. *South*/*North* see Section 3 (alternative calibration for *North*). The right panel displays the share of votes in favor of conservative fiscal policy in both countries. Y-axes are in percent.

C Further Results Country-Specific Fiscal Rules

Figure C.13: Impulse responses in *South* to a monetary policy shock with country-specific fiscal policy



Notes: Results conditional on *North* adopting a conservative fiscal policy ($\gamma^\tau \to \infty$). "Lib. + Cons." refers to the case when *South* adopts a liberal fiscal policy ($\gamma^\tau = 0.8$). "Cons. + Cons." refers to the case when *South* also adopts a conservative fiscal policy. *South*/*North* see Section 3. Y-axis: Percentage deviation from steady state, percentage points in case of inflation and interest rate. X-axis: Quarters.

D Details on the survey

The survey was conducted in May 2025 using Qualtrics, a professional service provider. The survey was originally designed by other researchers to track attitudes regarding wealth taxes and estate taxes, but we were allowed to include questions regarding the German debt brake. The group of respondents was selected to represent quotas of key demographic variables, including gender, age, income, and wealth (truncated within the survey form at zero, 74 respondents report zero). In total, data were collected from 1,088 respondents in Germany.

	Age	Household Income	Household Wealth
Mean	48.10	3815.83	97299.14
SD	18.05	3910.81	296343.7
Min	18	0	0
Max	92	39000	5000000

Table D.5: Descriptive Statistics. Outliers removed. Maximal age allowed = 100. All variables are self-reported.

First, the survey collected **demographic data** of the respondents. We use information from the following questions:

1. Please indicate your year of birth.
2. Please state your marital status.
3. How many people live in your household?

Second, the survey asked about **general political attitudes and the socio-economic status**. Here, we use the following questions

1. Are you affiliated with a political party? If yes, which party?
2. Please state your average monthly household income after taxes.
3. Please state your total household net worth in euros. Please include assets such as real estate, company shares, transportation, etc.

Finally, the survey asked the following questions regarding the **German debt brake** and the exemption to finance € 500 Billion additional infrastructure spending by issuing new debt

1. The debt brake was introduced in Germany to prevent the rise in government debt. Do you agree that the introduction of the debt brake was fundamentally a good idea?
2. Should there be exceptions to the debt brake for the financing of temporary infrastructure projects?

E A parsimonious representative-agent model to highlight the tax-timing trade-off

This appendix isolates the aggregate mechanism underlying the output trade-off discussed in Section 4.1. In a parsimonious representative-agent New Keynesian model, the timing of fiscal adjustment affects cumulative output through two opposing channels. For a given cumulative tax adjustment, front-loading is less contractionary because future tax increases depress current activity already in anticipation. At the same time, front-loading can increase the cumulative tax adjustment required to stabilize debt through a tax–inflation–real-rate feedback loop. We show that the relative strength of these two channels depends on the steady-state level of government debt.

Specifically, Lemma E.1 decomposes the output ranking into these timing and fiscal-burden channels. Proposition E.2 then shows that, beyond a threshold debt level, higher steady-state debt strengthens the latter and can eventually overturn the timing advantage of front-loaded adjustment. In our quantitative HANK model, this channel reduces but does not reverse the cumulative-output advantage of conservative fiscal policy for the debt levels of the two countries.

Setup. We consider the textbook representative-agent New Keynesian model with sticky prices, flexible wages and no capital, extended by fiscal policy which finances its expenditures through distortionary income taxes. We restrict the timing to three periods with adjustment in periods $t = 1, 2$ and a terminal steady state in period $t = 3$. We abstract from discounting by setting $\beta = 1$. The equilibrium is described by

$$\hat{y}_t = E_t \hat{y}_{t+1} - \left(\hat{i}_t - E_t \pi_{t+1}\right), \tag{47}$$

$$\pi_t = E_t \pi_{t+1} + \kappa \left(\chi \hat{y}_t + \tau_t\right), \tag{48}$$

$$\hat{i}_t = \phi_\pi \pi_t + u_t^m, \tag{49}$$

for $t = 1, 2$, where $\phi_\pi > 1$, $\kappa > 0$, and $\chi > 0$, plus a rule for the tax adjustment, τ_t, which enters marginal costs and, thus, the Phillips curve.

Let $\widetilde{\tau}_t$ denote the tax rate in levels and define

$$\tau_t \equiv -\log\left(\frac{1-\widetilde{\tau}_t}{1-\bar{\tau}}\right),$$

such that τ_t denotes the tax wedge entering marginal costs. We define $\widehat{y}_t = \log(y_t/\bar{y})$ and $\widehat{i}_t = \log(R_t/\bar{R})$. As in the quantitative model, the same tax rate applies to labor and profit income. Because we abstract here from capital, the taxable income equals output.

The government budget constraint is

$$d_t = \frac{R_{t-1}}{\Pi_t} d_{t-1} + \bar{g} - \widetilde{\tau}_t y_t,$$

with d_t denoting the debt level. Linearized around a zero-inflation steady state with $\bar{R} = 1$, we have

$$\widehat{d}_t = \widehat{d}_{t-1} + \widehat{i}_{t-1} - \pi_t - \frac{\bar{\tau}\bar{y}}{\bar{d}} \widehat{y}_t - \frac{(1-\bar{\tau})\bar{y}}{\bar{d}} \tau_t, \qquad t = 1, 2, 3 \tag{50}$$

with $\widehat{d}_t = (d_t - \bar{d})/\bar{d}$.

A contractionary monetary policy shock occurs in period 1:

$$u_1^m = \varepsilon_m > 0, \qquad u_2^m = 0.$$

Initially, the economy is in steady state, $\hat{d}_0 = \hat{i}_0 = 0$, and the terminal conditions are

$$\left(\hat{y}_3, \pi_3, \tau_3, \hat{d}_3\right) = (0, 0, 0, 0).$$

We compare two tax paths. Under conservative fiscal policy, the entire adjustment is front-loaded:

$$\tau_2^C = 0.$$

Under liberal fiscal policy, the adjustment is smoothed equally across periods:

$$\tau_1^L = \tau_2^L.$$

In either regime, taxes adjust so that the terminal debt condition $\hat{d}_3 = 0$ is satisfied. Define cumulative output and the cumulative tax adjustment as

$$\mathcal{Y}^r \equiv \hat{y}_1^r + \hat{y}_2^r, \qquad \Sigma^r \equiv \tau_1^r + \tau_2^r, \qquad r \in \{C, L\}.$$

A higher value of $\mathcal{Y}^r$ corresponds to a smaller cumulative output loss after a contractionary monetary policy shock.

Lemma E.1. *Let*

$$D \equiv 1 + \kappa\chi\phi_\pi.$$

For any tax path (τ_1, τ_2), cumulative output is

$$\mathcal{Y} = -\frac{\varepsilon_m}{D} - \frac{\kappa\phi_\pi}{D}\Sigma - \frac{\kappa(2\phi_\pi - 1)}{D^2}\tau_2. \tag{51}$$

Consequently, for a given cumulative tax adjustment Σ, front-loading reduces cumulative

output losses. Moreover, the difference between liberal and conservative fiscal policy can be written as

$$\mathcal{Y}^L - \mathcal{Y}^C = \underbrace{\frac{\kappa\phi_\pi}{D}\left(\Sigma^C - \Sigma^L\right)}_{\textit{fiscal-burden channel}} - \underbrace{\frac{\kappa(2\phi_\pi - 1)}{2D^2}\Sigma^L}_{\textit{timing channel}}. \tag{52}$$

The timing channel in (52) favors conservative fiscal policy: if both regimes require the same cumulative tax adjustment, then $\mathcal{Y}^C > \mathcal{Y}^L$. Liberal fiscal policy can generate a smaller cumulative output loss only if smoothing makes the required adjustment sufficiently smaller, $\Sigma^L < \Sigma^C$.

Proof. Solving the model backward gives

$$\hat{y}_2 = -\frac{\kappa\phi_\pi}{D}\tau_2, \tag{53}$$

$$\hat{y}_1 = -\frac{\varepsilon_m}{D} - \frac{\kappa\phi_\pi}{D}\tau_1 - \frac{\kappa(2\phi_\pi - 1)}{D^2}\tau_2. \tag{54}$$

Adding (53) and (54) yields (51). Under conservative fiscal policy, $\tau_2^C = 0$, whereas under liberal fiscal policy, $\tau_2^L = \Sigma^L/2$. Taking the difference between the two regimes therefore yields (52). □

To characterize the fiscal-burden channel, define

$$H \equiv 3 + \kappa\chi(\phi_\pi + 1)$$

and the debt threshold

$$\bar{d}^* \equiv \frac{(1-\bar{\tau})\bar{y}(2\phi_\pi - 1)}{\kappa(\phi_\pi - 1)^2}. \tag{55}$$

The regime-specific fiscal limits are

$$\bar{d}_C^{\max} \equiv \frac{(1-\bar{\tau})\bar{y}D - \kappa\phi_\pi\bar{\tau}\bar{y}}{\kappa(\phi_\pi - 1)}, \tag{56}$$

$$\bar{d}_L^{\max} \equiv \frac{2(1-\bar{\tau})\bar{y}D^2 - \kappa\bar{\tau}\bar{y}\left[\phi_\pi(4 + 2\kappa\chi\phi_\pi) - 1\right]}{\kappa(\phi_\pi - 1)H}. \tag{57}$$

As $\bar{d}$ approaches $\bar{d}_r^{\max}$ from below, the positive tax adjustment required under regime r diverges.

Proposition E.2. *Suppose*

$$\Xi \equiv \bar{y}\left\{\kappa(\phi_\pi - 1)\left[(1-\bar{\tau})\chi - \bar{\tau}\right] - (1-\bar{\tau})\right\} > 0. \tag{58}$$

Then

$$0 < \bar{d}^* < \bar{d}_C^{\max} < \bar{d}_L^{\max}.$$

For every debt level in the common admissible region, $0 < \bar{d} < \bar{d}_C^{\max}$, the difference in cumulative output is

$$\mathcal{Y}^L - \mathcal{Y}^C = \frac{\varepsilon_m}{H} \frac{\left[(1+\kappa\chi)\bar{d} + \bar{\tau}\bar{y}\right](\bar{d} - \bar{d}^*)}{(\bar{d}_C^{\max} - \bar{d})(\bar{d}_L^{\max} - \bar{d})}. \tag{59}$$

Hence, conservative fiscal policy produces the smaller cumulative output loss when $\bar{d} < \bar{d}^*$, whereas liberal fiscal policy produces the smaller cumulative output loss when $\bar{d}^* < \bar{d} < \bar{d}_C^{\max}$.

Thus, higher steady-state debt strengthens the fiscal-burden channel and can eventually overturn the timing advantage of front-loaded adjustment. Note that in our quantitative HANK model, the timing channel remains dominant in both countries, but its advantage is smaller in the higher-debt economy.

Proof. Combining the terminal debt condition with (47)–(50) yields the required cumulative tax adjustments

$$\Sigma^C(\bar{d}) = \frac{\left[(1+\kappa\chi)\bar{d} + \bar{\tau}\bar{y}\right]\varepsilon_m}{\kappa(\phi_\pi - 1)(\bar{d}_C^{\max} - \bar{d})}, \tag{60}$$

and

$$\Sigma^L(\bar{d}) = \frac{2D}{H} \frac{\left[(1+\kappa\chi)\bar{d} + \bar{\tau}\bar{y}\right]\varepsilon_m}{\kappa(\phi_\pi - 1)(\bar{d}_L^{\max} - \bar{d})}. \tag{61}$$

Substituting (60) and (61) into (52) and collecting terms gives (59).

Finally,

$$\bar{d}_C^{\max} - \bar{d}^* = \frac{\phi_\pi}{\kappa(\phi_\pi - 1)^2}\Xi, \tag{62}$$

and

$$\bar{d}_L^{\max} - \bar{d}_C^{\max} = \frac{D}{\kappa(\phi_\pi - 1)H}\Xi. \tag{63}$$

Under (58), both differences are positive. Since $\bar{d}^* > 0$ under the Taylor principle, this establishes the stated ordering. All factors in (59) are positive in the common admissible region except possibly $\bar{d} - \bar{d}^*$, which therefore determines the output ranking. □